

BSA's Contractual Representative	Overall Value Amount	Contract Type	Contract Number
L. Gonzales	TBD	Firm Fixed Price	477155

BROOKHAVEN NATIONAL LABORATORY

Brookhaven Science Associates

P.O. Box 5000

UPTON, L.I., N.Y. 11973-5000

Name and Address of Contractor: TBD

CONTRACT

This Contract (the "Contract") is between the party named above (the "Contractor"), and Brookhaven Science Associates, LLC ("BSA"), the latter acting under a Prime Contract with the United States of America (the "Government") represented by the United States Department of Energy ("DOE").

I. SCOPE OF WORK

The Contractor shall fabricate machined components for the Charge-Readout Plane (CRP) Patch Panel Mechanical Support and deliver them to Brookhaven Science Associates (BSA).

This contract includes a pre-production (qualification phase) requiring the Contractor to deliver twenty (20) sets of CRP Patch Panel Components. This contract also includes an option for a second production phase. At the sole discretion of BSA, and upon exercise of this option, the Contractor shall deliver an additional three hundred forty (340) sets of CRP Patch Panel Components. Exercise of the optional phase will be contingent upon successful completion and acceptance of the qualification phase.

All work shall be conducted in strict accordance with the following BSA exhibits attached hereto:

- Exhibit A - Statement of Work (SOW) entitled, "Charge-Readout Plane (CRP) Patch Panel Mechanical Support";
- Exhibit B – Technical Drawings; and
- Exhibit C – BNL Supplier Quality Assurance Requirements (BNL-QA-101).

- A. Reports/Deliverables:** The Contractor shall provide the following deliverable(s) to BSA's Technical Representative (TR):

Item	Description	Qty	Delivery Due Date	Requires BSA Approval
1	Phase 1: Pre-Production (Qualification Phase)	20	8 Weeks After Contract Award	Yes
2	Optional Phase 2: Production	43	Upon BSA's exercise of Optional Phase 2 + 1 Month	Yes
3	Optional Phase 2: Production	43	Upon BSA's exercise of Optional Phase 2 + 2 Months	Yes
4	Optional Phase 2: Production	43	Upon BSA's exercise of Optional Phase 2 + 3 Months	Yes
5	Optional Phase 2: Production	43	Upon BSA's exercise of Optional Phase 2 + 4 Months	Yes
6	Optional Phase 2: Production	43	Upon BSA's exercise of Optional Phase 2 + 5 Months	Yes
7	Optional Phase 2: Production	43	Upon BSA's exercise of Optional Phase 2 + 6 Months	Yes
8	Optional Phase 2: Production	43	Upon BSA's exercise of Optional Phase 2 + 7 Months	Yes
9	Optional Phase 2: Production	39	Upon BSA's exercise of Optional Phase 2 + 8 Months	Yes

II. PERIOD OF PERFORMANCE

This Contract shall be effective as of the date executed by both parties. The base period of performance shall consist of eight (8) weeks for completion and delivery of the Phase 1 (Pre-Production/Qualification Phase) deliverables, followed by a two (2) week evaluation and acceptance period by BSA.

BSA may, at its unilateral discretion, exercise one (1) optional phase (Phase 2: Production) for an additional period of eight (8) months. Exercise of the optional phase shall be contingent upon successful completion and acceptance of the Phase 1 deliverables. The option phase may only be exercised through a written modification at the prices set forth in Section III below. No work shall be performed under the option phase without a modification to this contract.

If the option is exercised, the total period of performance shall end ten (10) months and two (2) weeks from the date of award.

III. FIRM FIXED PRICE AND PAYMENT

A. **Firm Fixed Price:** In full consideration of the Contractor's performance hereunder, BSA shall pay the Contractor the firm fixed price of TBD U.S. Dollars (\$TBD). The said sum shall constitute full compensation for all services and materials furnished hereunder.

B. **Payment:** Payment will be made upon receipt and approval of properly certified invoices and upon receipt and approval of deliveries of CRP Patch Panel Components at the unit price listed below.

Description	Qty	Unit Price	Cumulative Toal
Phase 1: Pre-Production (Qualification Phase)	20		
Optional Phase 2: Production	340		

Payment terms are Net 10.

Invoices shall be directed to BSA's Accounts Payable Department via electronic or regular mail.

Electronic invoice submittal is preferred by BSA. The Contractor shall make every effort to have systems in place that allow for electronic invoice transmittal. Invoices shall be submitted in PDF format, via email, to apinvoices@bnl.gov.

Hard copy invoices, if applicable, shall be directed to BSA's Accounts Payable Department, Contracts Division, Bldg. No. 400D.

The Contractor shall indicate the final invoice by clearly marking such invoice as "FINAL." A copy of the final invoice must be submitted to BSA's Contractual Representative.

- C. **Freight:** Freight terms shall be *(for North American shipments insert:)* FOB BSA National Laboratory, 98 Rochester Street, Upton, Long Island, New York. Freight Prepaid, *(for foreign shipments insert:)* "DAP" Delivered At Place, BSA National Laboratory, 98 Rochester Street, Upton, Long Island, New York.

D. **Labelling, Packing, and Notification Requirements:**

More detailed instructions are available at <https://www.bnl.gov/shipping/> or by emailing WarehouseOperations@bnl.gov.

Labeling Requirements

All shipments must comply with the following labelling requirements. Failure to do so may result in delayed payments. Labels must be placed on the upper left portion of the container when possible and contain:

Brookhaven National Laboratory
Brookhaven's Contract Number
98 Rochester St.
Upton, NY 11973
Box Number / Number of Boxes

Packing List Requirements

Every shipping container must include a packing list. The packing list must be clearly identified and affixed to a side of the shipping container. At a minimum, the packing list should contain the following information:

- Ship from company name and address

- Supplier contact name
- Supplier telephone number
- Ship to address as instructed by Buyer/Contracts Specialist
- Brookhaven's Contract number
- Part number, description, quantity, unit of measure, and corresponding BNL Contract line number

Special Notification of Delivery

If a shipment is heavier than 3,000 lbs., longer than 8 feet in any direction, or has a high center of gravity, it will require rigging upon receipt and the shipper shall provide a minimum of 3 days' advance notice of delivery to BNL's Receiving Department via email to WarehouseOperations@bnl.gov. The shipper must obtain acknowledgement from BNL that delivery notification was received. Failure to contact BNL will result in delays during delivery and any resulting demurrage will be at the shipper's expense.

Marking of Outer Package

Each shipping container (transport package) with a gross weight equal to or greater than 400 lbs. (180 kilograms) must be marked with the center of gravity location and gross weight on at least one side or end panel. It must also meet the requirements of the latest revision of [ASTM Standard D 3951](#), Standard Practice for Commercial Packaging. This applies to any item shipped (originally packaged or repackaged) to BNL or other locations cited in the Contract.

Protection of Material and Equipment

The Contractor shall employ procedures which ensure adequate protection of material and equipment during shipment and while in storage. Such protection shall include special environmental packaging as necessary. All items shipped (originally packaged or repackaged) to BNL or other locations cited in the PO or Contract shall comply with the requirements for preservation, packaging, and marking as stated in the latest revision of [ASTM Standard D 3951-10](#), Standard Practice for Commercial Packaging.

Unacceptable Packing Material

Do not use vermiculite as a packaging material inside any shipments to BNL unless required by shipping regulation. "Vermiculite" as defined in this specification means the hydrous, silicate mineral classified as a phyllosilicate that has been expanded by heating. Vermiculite used as a packaging material in shipments must be determined to be free of asbestos (<1% chrysolite, amosite, crocidolite, tremolite, anthophyllite, and actinolite). All shipments containing vermiculite must be accompanied by documentation stating that the vermiculite used is certified asbestos-free. This documentation must be affixed to the outside of the package.

Radioactive Shipments

Shipments to Brookhaven National Laboratory (BNL) containing radioactive material require pre-authorization prior to shipment. Contact the following staff in the Packaging & Transportation Group for instructions:

- Brett Fagan, Ext. 5645, bfagan@bnl.gov
- Kenneth Hansen, khansen2@bnl.gov
- Steven Woodburn, Ext. 3271, woodburn@bnl.gov

Ocean Shipments

Foreign shipments transported via ocean freight must use the [10+2 REQUIREMENTS Form](#) (PDF). The form must be completed by the Contractor 72 hours before departure of the vessel. Send form to the following:

Interfreight Harmonized Logistics, Inc.

Fax: 516-371-6880

bnl@interfreight.net

Brookhaven National Laboratory

WarehouseOperations@bnl.gov

Once the form is submitted to customs, an acceptance number will be provided. Freight will not be loaded onto a vessel without an acceptance number. Failure to submit the 10+2 form may result in fines. It is the Contractor's responsibility to make sure the 10+2 form is submitted 72 hours prior to the vessel sailing. BSA shall withhold payment from the Contractor's final invoice for any penalties incurred by BSA due to late or incorrect ISF filing by the Contractor.

IV. AUTHORIZED REPRESENTATIVES

- A. **BSA's Technical Representative:** _____, telephone no. (631) 344-_____, _____@bnl.gov (email), is BSA's Technical Representative, and shall act as liaison between BSA and the Contractor in technical matters only.
- B. **BSA's Contractual Representative:** Lisa Gonzales, telephone no. (631) 344-5965, lgonzales@bnl.gov (email), is BSA's Contractual Representative. Any change or modification in the terms and conditions of this Contract shall require the written approval of BSA's Contractual Representative or designee.
- C. **Contractor's Technical Representative:** _____, telephone no. _____, _____ (email), is the Contractor's authorized Technical Representative, and shall act as point of contact between BSA and the Contractor in technical matters only.
- C. **Contractor's Contractual Representative:** _____, telephone no. _____, _____ (email), is the Contractor's authorized Contractual Representative and shall act as liaison between BSA and the Contractor.

IV. ADDITIONAL TERMS

The provisions of Brookhaven Science Associates, LLC, General Terms and Conditions for Non-Commercial Items, (*Rev 24.0, April 2026*) are incorporated herein and made a part hereof.

This Contract is issued pursuant to Brookhaven Science Associates General and/or Supplemental Terms and Conditions listed above, which are incorporated herein by reference. The complete text of these terms and conditions can be viewed at [PPM | Terms and Conditions](#). Should the Contractor have any questions and/or not have access to these terms and conditions, contact the Contracts Specialist immediately.

The following documents attached hereto and made a part hereof, contain additional provisions of this Contract:

- Exhibit A - Statement of Work (SOW) entitled, “Charge-Readout Plane (CRP) Patch Panel Mechanical Support”;
- Exhibit B – Technical Drawings; and
- Exhibit C – BNL Supplier Quality Assurance Requirements (BNL-QA-101).

This Contract does not bind nor purport to bind the Government of the United States.

ACCEPTED:

CONTRACTOR

BROOKHAVEN SCIENCE ASSOCIATES, LLC.

By: _____

By: _____

Name: _____

Name: Lisa Gonzales

Title: _____

Title: Subcontracts Supervisor

Date: _____

Date: _____

The Contractor shall sign and return one (1) copy of the Contract to BSA’s Contractual Representative, Lisa Gonzales, Procurement and Property Management Division.

Exhibit A



P.O. Box 5000
Upton, NY 11973-5000

managed by Brookhaven Science Associates
for the U.S. Department of Energy

2026/03/17

Subject: Statement of Work for Charge-Readout Plane (CRP) Patch Panel

Mechanical Support

1. Fabrication Scope

The vendor shall fabricate and deliver machined components for the **CRP Patch Panel Mechanical Support** in accordance with the following documentation:

- **Parts List:** *List of machined parts for CRP Patch Panel Mechanical Support* (detailing parts and quantities per set).

List of machined parts for CRP Patch Panel Mechanical Support

ITEM	QTY PER SET	DRAWING NUMBER	PART NAME / DESCRIPTION	REVISION
1	1	DCVDCRPPPV20103	ANGLE-CLAMP_BAR-MINISAS-LEFT	-
2	1	DCVDCRPPPV20102	ANGLE-CLAMP_BAR-MINISAS-RIGHT	-
3	1	DCVDCRPPP014	CLAMP_BAR-MINISAS-PATCH_PANEL	D
4	1	DCVDCRPPPV2015	CLMP_BAR-EDGE CARD-PATCH_PANEL_VD	A
5	2	DCVDCRPPPV203	PATCH_PANEL_BRACKET_SPACER-G10	-
6	6	DCVDCRPPP09	PP_CLAMP_BAR_DATA_CABLE	A
7	6	DCVDCRPPPV209	PP_PLATE_NUT_DATA_CABLE	A
8	6	DCVDCRPPP010	PP_PWR_CABLE_CLAMP_HALF	-
9	6	DCVDCRPPPV2010	PP_PWR_CABLE_CLAMP_HALF_CLEAR	-
10	2	DCVDCRPPPV20201	TUBE_SQ-CRP_PP_FIXED-EDGE_CARD	-
11	2	DCVDCRPPPV20101	TUBE_SQ-CRP_PP_FIXED-MINISAS	A

- **Drawings:** *CRP Patch Panel Mechanical Support_drawing.zip*.

2. Materials and Fabrication Requirements

The Vendor shall ensure each fabricated/machined part meets the following requirements:

Materials

- Aluminum parts: 6061-T6 / T6511 per ASTM B221 or equivalent

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Export Administration Regulations 15 CFR Parts 730-774 as EAR99 Technology.*

- Stainless steel parts: 304 / 304L per ASTM A276 / A554
- G10 parts: per MIL-I-24768/27

Surface Treatment

- Aluminum parts shall receive chromate conversion coating (MIL-DTL-5541, Class 3, clear, RoHS compliant) unless otherwise specified.

Machining and Tolerances

- All machining must comply with the tolerances specified on the individual drawings.
- Dimensions and tolerances shall follow the drawings and 3D models.
- Untoleranced dimensions are basic unless otherwise specified.

Edge and Finish Requirements

- Break all sharp edges: 0.12 mm min to 0.75 mm max
- Surface finish: 3.2 μm (125 μin) or better, unless otherwise specified.

3. Identification and Marking

The Vendor shall ensure each fabricated/machined part is marked in accordance with the following requirements.

- Each part shall be permanently marked with:
 - Drawing number
 - Revision letter
- Marking shall comply with MIL-STD-130
- Minimum character height:
 - Aluminum parts: $\geq 1.2\text{--}1.6$ mm (per drawing)
 - Stainless steel / G10 parts: ≥ 3 mm
- Marking location: approximate location per drawing

4. Cleaning and Packaging

The Vendor shall ensure the fabricated/machined parts are cleaned and packed in accordance with the following requirements.

- All parts shall be:
 - Free of oils, grease, and contaminants
- Cleaning methods:
 - Detergent and alcohol cleaning permitted
- Packaging:
 - Individual or grouped packaging to prevent damage
 - Maintain cleanliness during shipment

5. Quality and Verification

- Vendor shall ensure all parts:
 - Meet dimensional and material specifications
 - Conform to drawings and referenced standards
- No burrs, sharp edges, or machining defects permitted
- Upon request from BSA, Vendor shall provide the following:
 - Certificate of Conformance (CoC)
 - Material certifications
- BNL reserves the right to:
 - Inspect parts upon receipt
 - Reject non-conforming parts

6. Fabrication Phases

This procurement will be executed in two phases:

Phase 1 – Pre-Production (Qualification Phase)

- Quantity: **20 sets**
- Purpose: vendor qualification and design verification

- Delivery: within **8 weeks** after PO

Vendor shall:

- Deliver complete sets (all parts per set)

Phase 2 – Production (Optional)

- Quantity: **340 sets**
- Initiation: Phase 2 will begin upon written modification to contract exercising Option 1, Phase 2.

Requirements:

- Delivery within **8 months** after authorization
- Batch delivery:
 - Monthly or other schedule approved by BSA Technical Representative. (e.g., 40–50 sets/month)

Signature: zhao Mankong 03/17/2026
Of Requestor or Technical Rep

* * *

4

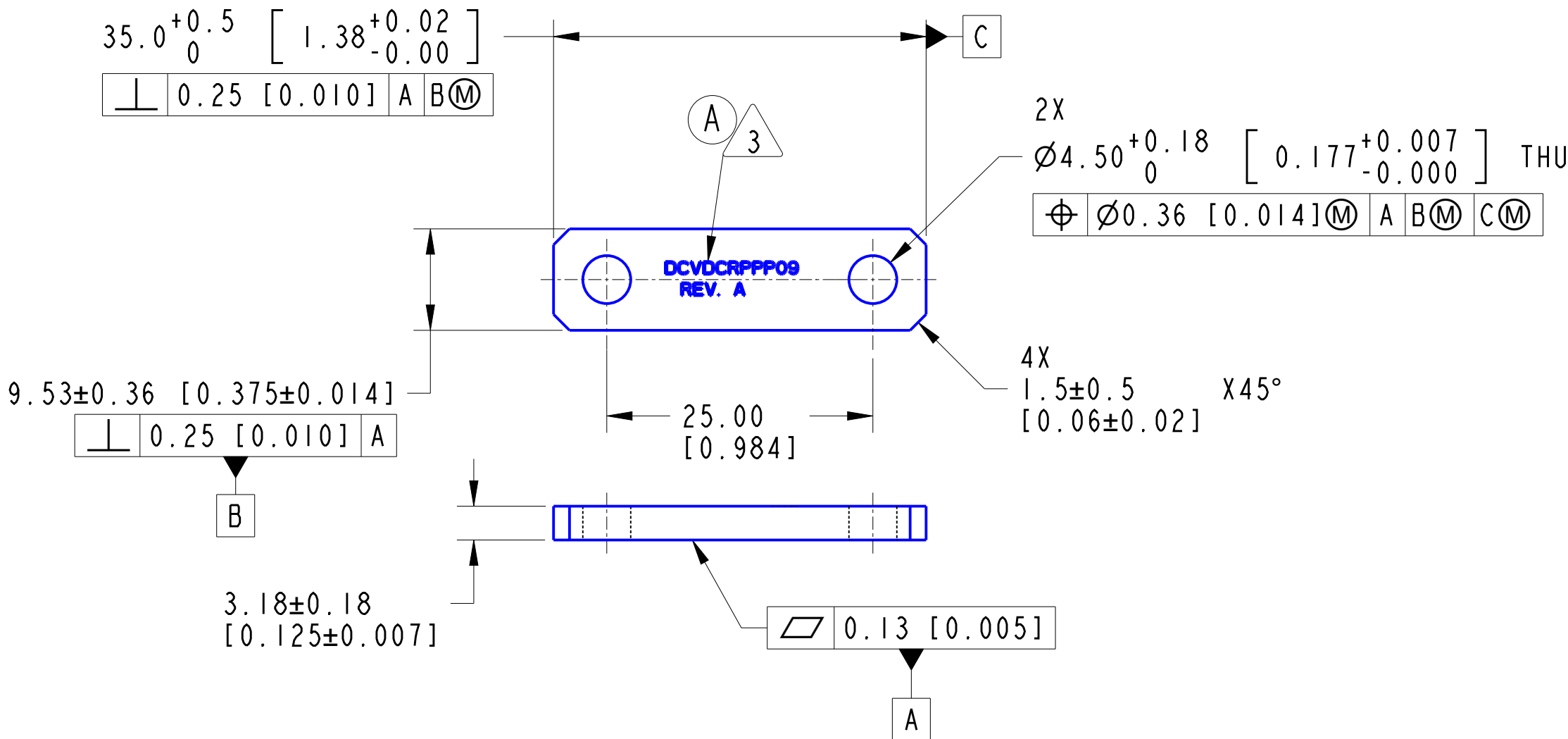
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Exhibit B

REVISIONS					
REV	ECN NO.	DESCRIPTION	BY	DATE	CKR
-		INITIAL RELEASE.	JF	8/8/2022	MZ
A		NO PHYSICAL CHANGE; ESH&Q RISK LEVEL WAS A-4; ADDED EXPORT CONTROL NOTE ABOVE BOM	JF	2/13/2026	MZ



ISOMETRIC VIEW
FOR REFERENCE ONLY

5	PART TO BE FREE OF OILS, GREASE, ETC. PERMISSIBLE TO USE DETERGENTS AND ALCOHOL FOR CLEANING. PART PACKAGING TO MAINTAIN CLEANLINESS.
4	THIS DRAWING MAY BE USED WITH MODEL "PP_CLAMP_BAR_DATA_CABLE REV.A.stp" FOR COMPLETE PRODUCT DEFINITION. UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC. VALUES QUERIED FROM 3-D DIGITAL DATA FILE ARE BASIC.
3	ETCH PART WITH DRAWING NUMBER & REV. LETTER WITH MIN. 1.2 mm HIGH CHARACTERS IN ACCORDANCE WITH MIL-STD-130. LOCATE APPROXIMATELY WHERE SHOWN.
2	ALUMINUM CHROMATE per MIL-DTL-5541, CLASS 3, CLEAR (RoHS COMPLIANT) Iridite, Alodine, or EQUIVALENT.
1	MATERIAL: 1/8" THK. X 1/2" WIDE ALUMINUM SHEET or BAR 6061-T6511, ASTM B221, SAE AMS QQ-A-200/8 Suggested Source: McMaster PN: 8975K577
	Notes: Unless otherwise specified:

Do Not Scale Drawing

DCVDCRPPV201	6
DCVDCRPPP01	12
NEXT ASSEMBLY	QTY.
APPLICATION	

DIMENSIONING & TOLERANCING IN ACCORDANCE WITH ASME Y14.5-2009 UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC DIMENSIONS ARE IN MILLIMETERS DIMENSIONS IN BRACKETS [x.xx] ARE IN INCHES AND ARE FOR REFERENCE ONLY		Brookhaven National Laboratory Physics Department / High-Energy Physics		DUNE DEEP UNDERGROUND NEUTRINO EXPERIMENT	
DRAWN BY J. FARRELL		Aug-03-22		Cold Electronics	
CHECKED BY M. ZHAO		8/3/2022		TOP ASSEMBLY: CRP_BOTTOM_PATCH_PANEL_INSTLN	
ENGINEER M. ZHAO		8/3/2022		NEXT ASSEMBLY: FD2-VD_CRP_PATCH_PANEL_ASSY	
ENGR MGR				DESCRIPTION: PP_CLAMP_BAR_DATA_CABLE	
CHIEF M.E.				SIZE: C	
Q.A. G. MUSSI		2/13/2026		DRAWING PART NUMBER: DCVDCRPPP09	
WBS#				ESH&Q RISK LEVEL A-3	
BREAK EDGES & SHARP CORNERS 0.12 [0.005"] MIN. TO 0.75 [0.030"] MAX.		FINISH 3.2 [125]		WEIGHT: kg 0.00	
THIRD ANGLE PROJECTION		MODEL NAME: PP_CLAMP_BAR_DATA_CABLE		WEIGHT: lbs. 0.006	
SCALE: 2:1				SHEET: 1/1	
				REV. A	

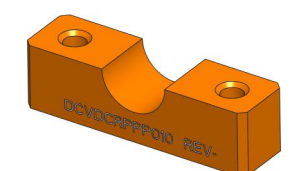
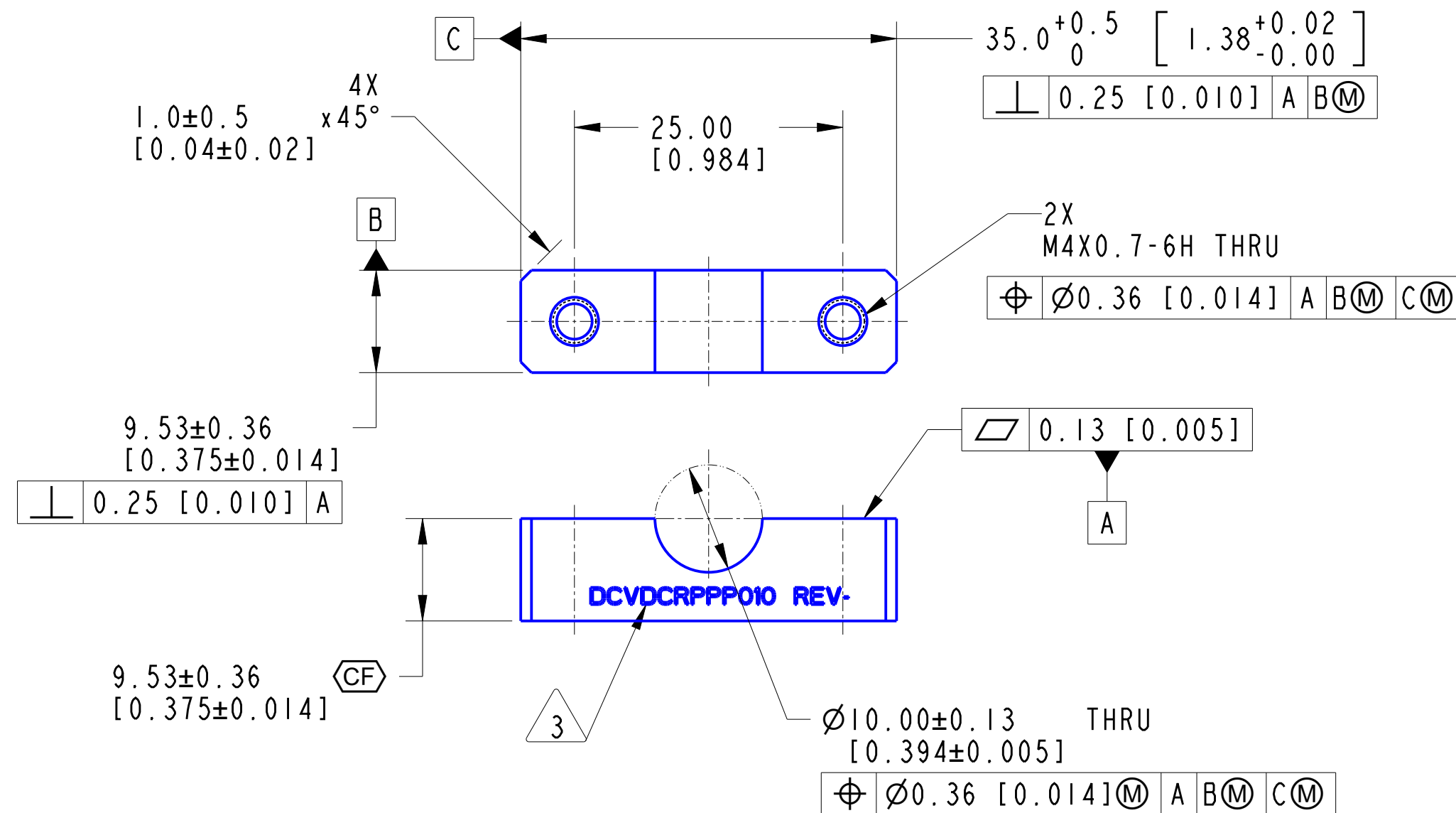
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1

REVISIONS						
REV	ECN NO.	DESCRIPTION	BY	DATE	CKR	APP
-		INITIAL RELEASE.	JF	8/8/2022	MZ	MZ



ISOMETRIC VIEWS
FOR REFERENCE ONLY

This information is controlled by the U.S. Department of Commerce
Export Administration Regulations 15 CFR Parts 730-774 as EAR99 Technology.

UNTOLERANCED DIMENSIONS ARE BASIC

Do Not Scale Drawing

5	PART TO BE FREE OF OILS, GREASE, ETC. PERMISSIBLE TO USE DETERGENTS AND ALCOHOL FOR CLEANING. PART PACKAGING TO MAINTAIN CLEANLINESS.
4	THIS DRAWING MAY BE USED WITH MODEL "PP_PWR_CABLE_CLAMP_HALF REV-.stp" FOR COMPLETE PRODUCT DEFINITION. UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC. VALUES QUERIED FROM 3-D DIGITAL DATA FILE ARE BASIC.
3	ETCH PART WITH DRAWING NUMBER & REV. LETTER WITH MIN. 1.6 mm HIGH CHARACTERS IN ACCORDANCE WITH MIL-STD-130. LOCATE APPROXIMATELY WHERE SHOWN.
2	ALUMINUM CHROMATE per MIL-DTL-5541, CLASS 3, CLEAR (RoHS COMPLIANT) Iridite, Alodine, or EQUIVALENT.
1	MATERIAL: 0.375" THK. ALUM. 6061-T6511 SHEET or BAR per SAE AMS QQ-A-200/8. Suggested Source: McMaster PN: 9008K78
	Notes: Unless otherwise specified:

DCVDCRPPP201	6
DCF2-VD_CRP_PP	12
NEXT ASSEMBLY	QTY.
APPLICATION	

DIMENSIONING & TOLERANCING IN ACCORDANCE WITH ASME Y14.5-2009		 Brookhaven National Laboratory		Physics Department / High-Energy Physics		 DEEP UNDERGROUND NEUTRINO EXPERIMENT											
UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC DIMENSIONS ARE IN MILLIMETERS																	
DIMENSIONS IN BRACKETS [x.xx] ARE IN INCHES AND ARE FOR REFERENCE ONLY		DRAWN BY		J. FARRELL		Aug-02-22		Cold Electronics									
<i>DIMENSIONAL TOLERANCES</i>		CHECKED BY		M. ZHAO		8/29/2022		TOP ASSEMBLY: CRP_BOTTOM_PATCH_PANEL_INSTLN									
		ENGINEER		M. ZHAO		8/29/2022		NEXT ASSEMBLY: FD2-VD_CRP_PATCH_PANEL_ASSY									
		ENGR MGR						DESCRIPTION: PP_PWR_CABLE_CLAMP_HALF									
		CHIEF M.E.						SIZE: C		DRAWING PART NUMBER: DCVDCRPPP010							
BREAK EDGES & SHARP CORNERS 0.12 [0.005"] MIN. TO 0.75 [0.030"] MAX.		FINISH 3.2 [125] ✓		Q.A.		J. PAUCAR		5/6/2025									
 THIRD ANGLE PROJECTION		MODEL NAME: PP_PWR_CABLE_CLAMP_HALF		WBS#						ESH&Q RISK LEVEL: A-3		WEIGHT: 0.01 kg		SHEET: 1/1		REV. -	
										SCALE: 2:1		WEIGHT: 0.016 lbs					

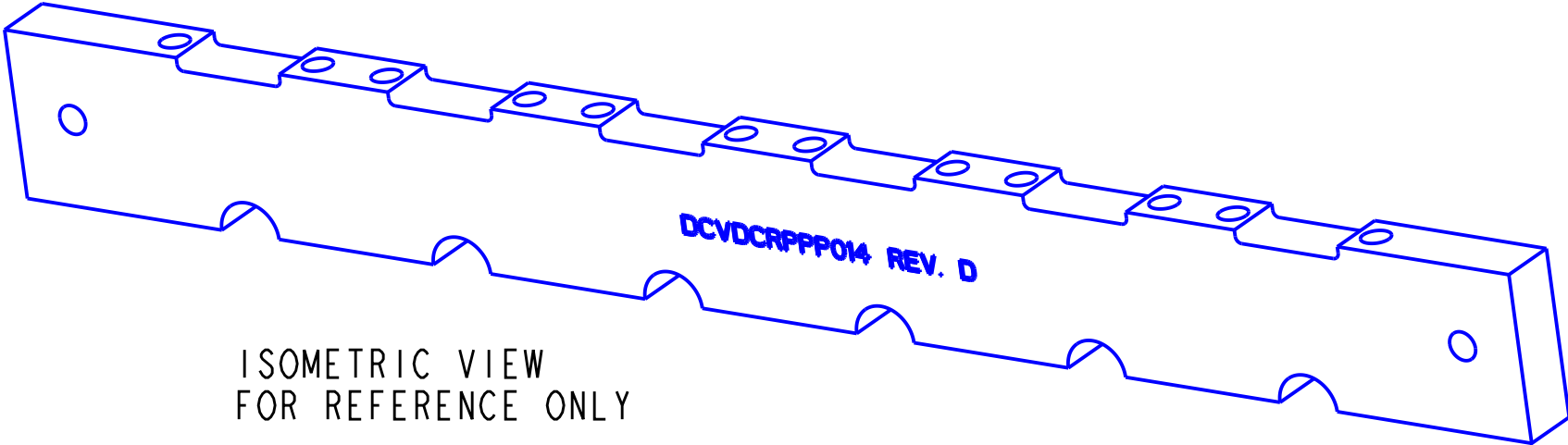
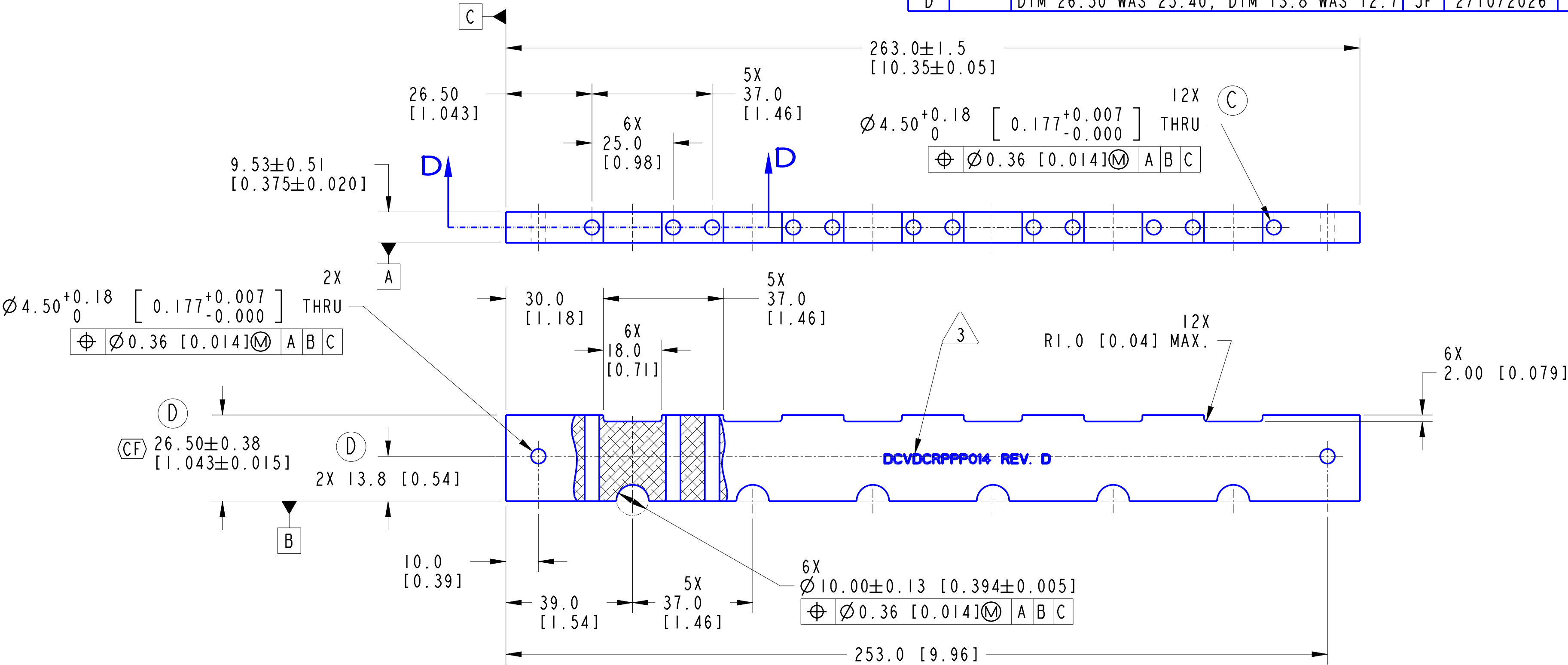
4

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2

1

REVISIONS						
REV	ECN NO.	DESCRIPTION	BY	DATE	CKR	APP
B		INITIAL RELEASE "B" TO MATCH ASSY REV.	JF	11/4/2022	MZ	MZ
C		12X M4 CLEARANCES HOLES WERE TAPPED M4.	JF	11/14/2022	MZ	MZ
D		DIM 26.50 WAS 25.40; DIM 13.8 WAS 12.7	JF	2/10/2026	MZ	MZ



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Export Administration Regulations 15 CFR Parts 730-774 as EAR99 Technology.

UNTOLERANCED DIMENSIONS ARE BASIC
DIMENSIONAL TOLERANCES

	0.25 [0.010]	A	B	C
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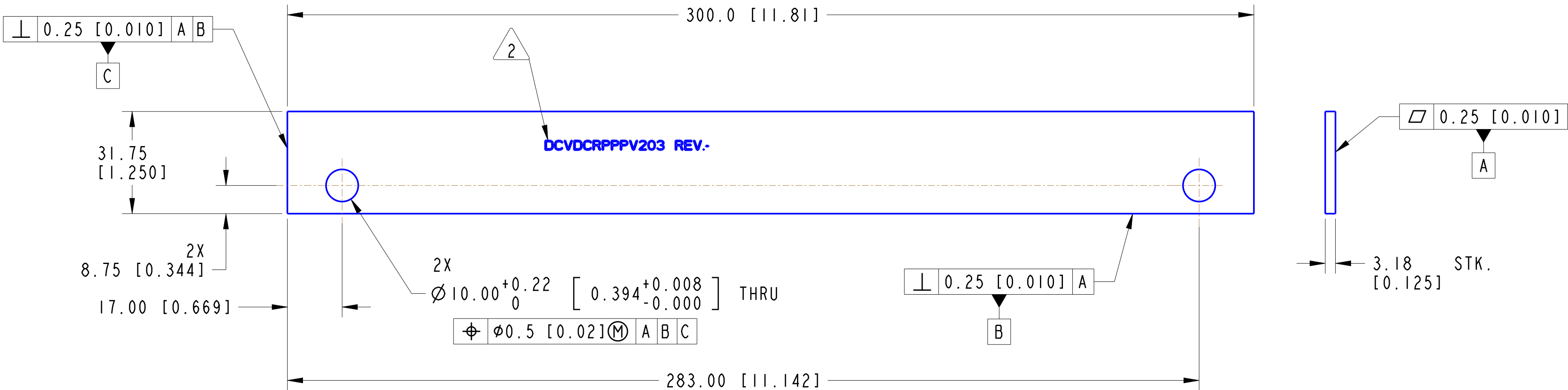
Do Not Scale Drawing

5	THIS DRAWING MAY BE USED WITH MODEL "CLAMP_BAR-MINISAS-PATCH_PANEL_REV.D.stp" FOR COMPLETE PRODUCT DEFINITION. UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC. VALUES QUERIED FROM 3-D DIGITAL DATA FILE ARE BASIC.
4	PART TO BE FREE OF OILS, GREASE, ETC. PERMISSIBLE TO USE DETERGENTS AND ALCOHOL FOR CLEANING. PART PACKAGING TO MAINTAIN CLEANLINESS.
3	ETCH PART WITH DRAWING NUMBER & REV. LETTER WITH MIN. OF 1.59mm HIGH CHARACTERS IN ACCORDANCE WITH MIL-STD-130. LOCATE APPROXIMATELY WHERE SHOWN.
2	ALUMINUM CHROMATE per MIL-DTL-5541, CLASS 3, CLEAR (RoHS COMPLIANT) Iridite, Alodine, or EQUIVALENT.
1	MATERIAL: 6061-T6 ALUMINUM BAR per ASTM B221, Stock Size: 3/8" Thick x 1" Wide x 12" Lg Suggested Source: McMaster PN: 8975K54
	Notes: Unless otherwise specified:

DCVDCRPPPV201	I
DCVDCRPPPO1	I
NEXT ASSEMBLY	QTY.
APPLICATION	

DIMENSIONING & TOLERANCING IN ACCORDANCE WITH ASME Y14.5-2009		Brookhaven National Laboratory Physics Department / High-Energy Physics		DUNE DEEP UNDERGROUND NEUTRINO EXPERIMENT	
UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC. DIMENSIONS ARE IN MILLIMETERS.		DRAWN BY	J. FARRELL	Oct-26-22	Cold Electronics FD2-VD_PATCH_PANEL_ASSY_ON_CRP CRP_PATCH_PANEL_FIXED-MINISAS CLAMP_BAR-MINISAS-PATCH_PANEL
DIMENSIONS IN BRACKETS [x.xx] ARE IN INCHES AND ARE FOR REFERENCE ONLY		CHECKED BY	M. ZHAO	10/26/2022	
DIMENSIONAL TOLERANCES		ENGINEER	M. ZHAO	10/26/2022	
BREAK EDGES & SHARP CORNERS 0.12 [0.005"] MIN. TO 0.75 [0.030"] MAX.		ENGR MGR			
THIRD ANGLE PROJECTION		CHIEF M.E.			DCVDCRPPPO14
FINISH 3.2 [125]		Q.A.	J. PAUCAR	5/6/2025	
MODEL NAME: CLAMP_BAR-MINISAS-PATCH_PANEL		WBS#			
		ESH&Q RISK LEVEL	A-3	WEIGHT: kg 0.15	REV. D
		SCALE: 1:1		WEIGHT: lbs. 0.339	

REVISIONS					
REV	ECN NO.	DESCRIPTION	BY	DATE	CKR APP
-		INITIAL RELEASE	JF	3/7/2025	MZ MZ



3	PART TO BE FREE OF OILS, GREASE, ETC. PERMISSIBLE TO USE DETERGENTS AND ALCOHOL FOR CLEANING. PART PACKAGING TO MAINTAIN CLEANLINESS.
2	MARK PART WITH DRAWING NUMBER & REV. LETTER WITH MIN. OF 3mm HIGH CHARACTERS IN ACCORDANCE WITH MIL-STD-130. LOCATE APPROXIMATELY WHERE SHOWN.
1	MATERIAL: G-10/FR4 SHEET/BAR per Mil-I-24768/27 Suggested Sources: www.mcmaster.com/8557K12-8557K122 www.professionalplastics.com
	Notes: Unless otherwise specified:

Do Not Scale Drawing

This information is controlled by the U.S. Department of Commerce
Export Administration Regulations 15 CFR Parts 730-774 as EAR99 Technology.

DIMENSIONING & TOLERANCING IN ACCORDANCE WITH ASME Y14.5-2009 UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC DIMENSIONS ARE IN MILLIMETERS DIMENSIONS IN BRACKETS [x.xx] ARE IN INCHES AND ARE FOR REFERENCE ONLY		Brookhaven National Laboratory Physics Department / High-Energy Physics		DEEP UNDERGROUND NEUTRINO EXPERIMENT	
DRAWN BY		J. FARRELL	Mar-07-25	Cold Electronics	
CHECKED BY		M. ZHAO	3/7/2025	TOP ASSEMBLY: FD2-BDE-CRP-010101/-010201	
ENGINEER		M. ZHAO	3/7/2025	NEXT ASSEMBLY: CRP_PATCH_PANEL_FIXED	
ENGR MGR				DESCRIPTION: PATCH_PANEL_BRACKET_SPACER-G10	
CHIEF M.E.				SIZE: C	
Q.A.		J. PAUCAR	4/30/2025	DRAWING PART NUMBER: DCVDCRPPPV203	
WBS#				ESH&Q RISK LEVEL A-3	
BREAK EDGES & SHARP CORNERS 0.12 [0.005"] MIN. TO 0.75 [0.030"] MAX.		FINISH 3.2 [125]		WEIGHT: kg 0.05	SHEET: 1/1
THIRD ANGLE PROJECTION		MODEL NAME: PATCH_PANEL_BRACKET_SPACER-G10		WEIGHT: lbs. 0.118	REV: -

D

C

B

A

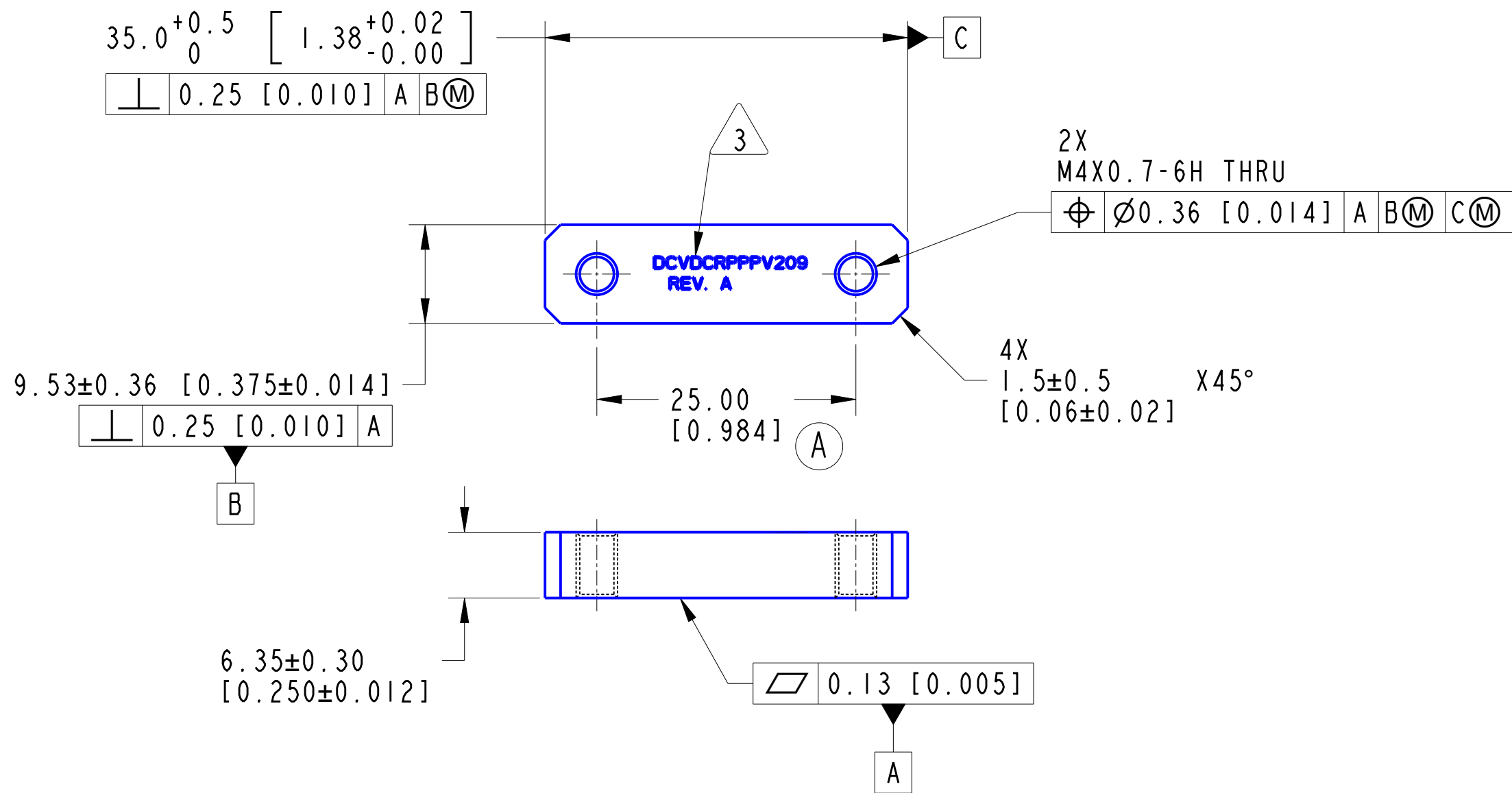
D

C

B

A

REVISIONS						
REV	ECN NO.	DESCRIPTION	BY	DATE	CKR	APP
-		INITIAL RELEASE.	JF	4/28/2025	MZ	MZ
A		ADDED DIMENSION 25.00 IN ZONE C2	JF	8/11/2025	MZ	MZ



ISOMETRIC VIEW
FOR REFERENCE ONLY

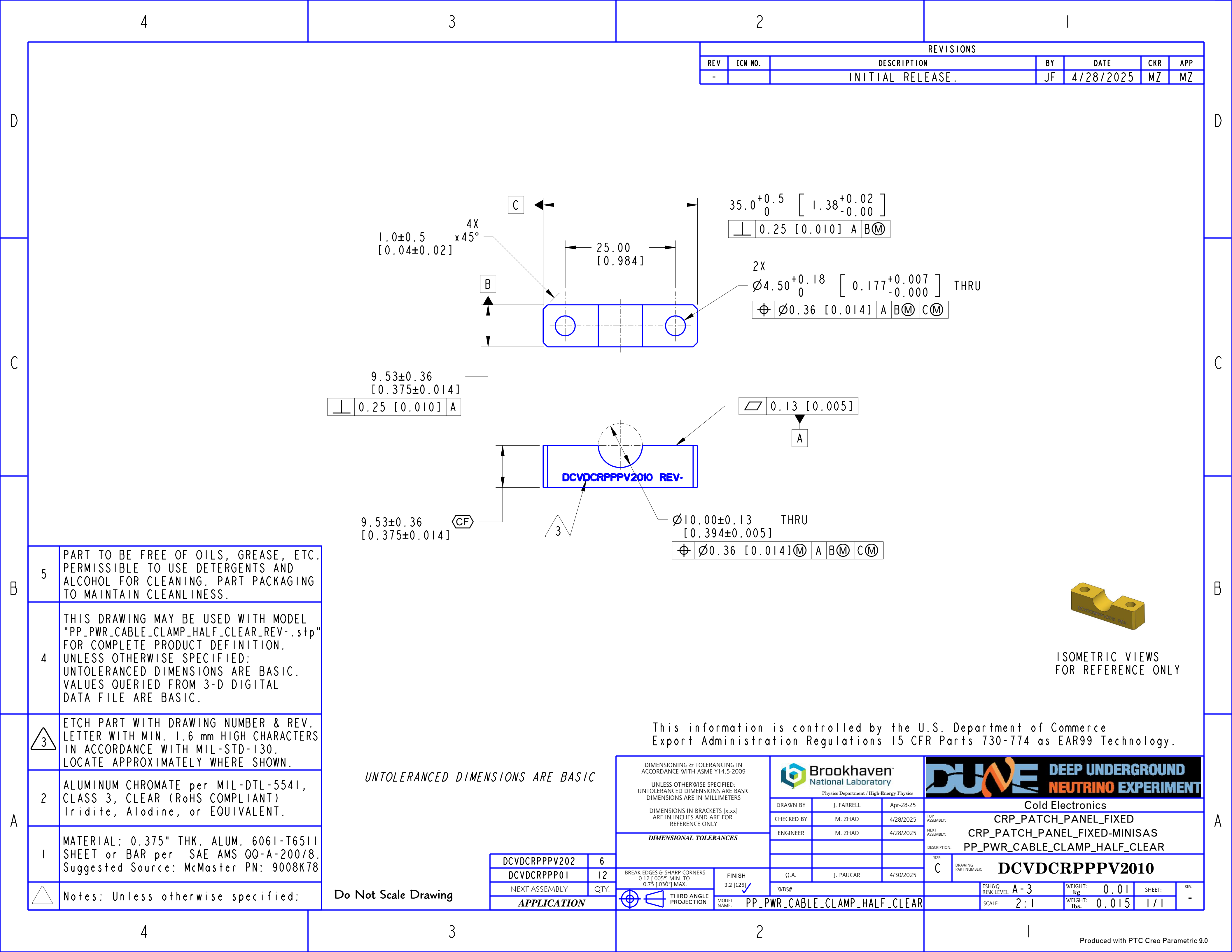
This information is controlled by the U.S. Department of Commerce
Export Administration Regulations 15 CFR Parts 730-774 as EAR99 Technology.

5	PART TO BE FREE OF OILS, GREASE, ETC. PERMISSIBLE TO USE DETERGENTS AND ALCOHOL FOR CLEANING. PART PACKAGING TO MAINTAIN CLEANLINESS.
4	THIS DRAWING MAY BE USED WITH MODEL "PP_PLATE_NUT_DATA_CABLE_REV_A.stp" FOR COMPLETE PRODUCT DEFINITION. UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC. VALUES QUERIED FROM 3-D DIGITAL DATA FILE ARE BASIC.
3	ETCH PART WITH DRAWING NUMBER & REV. LETTER WITH MIN. 1.2 mm HIGH CHARACTERS IN ACCORDANCE WITH MIL-STD-130. LOCATE APPROXIMATELY WHERE SHOWN.
2	ALUMINUM CHROMATE per MIL-DTL-5541, CLASS 3, CLEAR (RoHS COMPLIANT) Iridite, Alodine, or EQUIVALENT.
1	MATERIAL: 1/4" THK. X 3/8" WIDE ALUMINUM SHEET or BAR 6061-T6511, ASTM B221, SAE AMS QQ-A-200/8 Suggested Source: McMaster PN: 8975K589
	Notes: Unless otherwise specified:

Do Not Scale Drawing

DCVDCRPPV201	12
NEXT ASSEMBLY	QTY.
APPLICATION	

DIMENSIONING & TOLERANCING IN ACCORDANCE WITH ASME Y14.5-2009 UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC DIMENSIONS ARE IN MILLIMETERS DIMENSIONS IN BRACKETS [x.xx] ARE IN INCHES AND ARE FOR REFERENCE ONLY		Brookhaven National Laboratory Physics Department / High-Energy Physics		DUNE DEEP UNDERGROUND NEUTRINO EXPERIMENT	
DRAWN BY J. FARRELL		Apr-28-25		Cold Electronics	
CHECKED BY M. ZHAO		4/28/2025		TOP ASSEMBLY: CRP_PATCH_PANEL_FIXED	
ENGINEER M. ZHAO		4/28/2025		NEXT ASSEMBLY: CRP_PATCH_PANEL_FIXED-MINISAS	
				DESCRIPTION: PP_PLATE_NUT_DATA_CABLE	
				SIZE: C	
				DRAWING PART NUMBER: DCVDCRPPV209	
BREAK EDGES & SHARP CORNERS 0.12 [0.005"] MIN. TO 0.75 [0.030"] MAX.		FINISH 3.2 [125]		ESH&Q RISK LEVEL A-3	
THIRD ANGLE PROJECTION		Q.A. J. PAUCAR 4/30/2025		WEIGHT: 0.01 kg	
MODEL NAME: PP_PLATE_NUT_DATA_CABLE		WBS#		WEIGHT: 0.012 lbs.	
				SCALE: 2:1	
				SHEET: 1/1	
				REV. A	



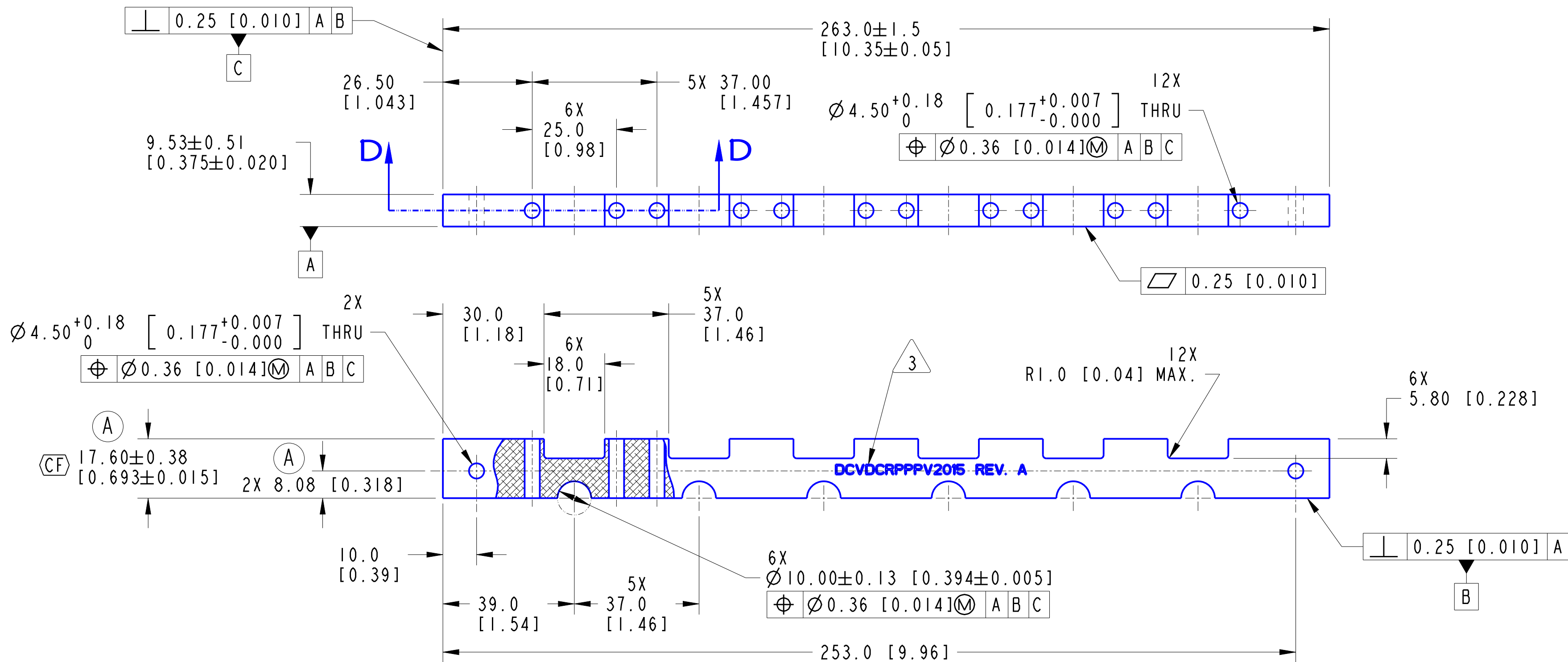
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3

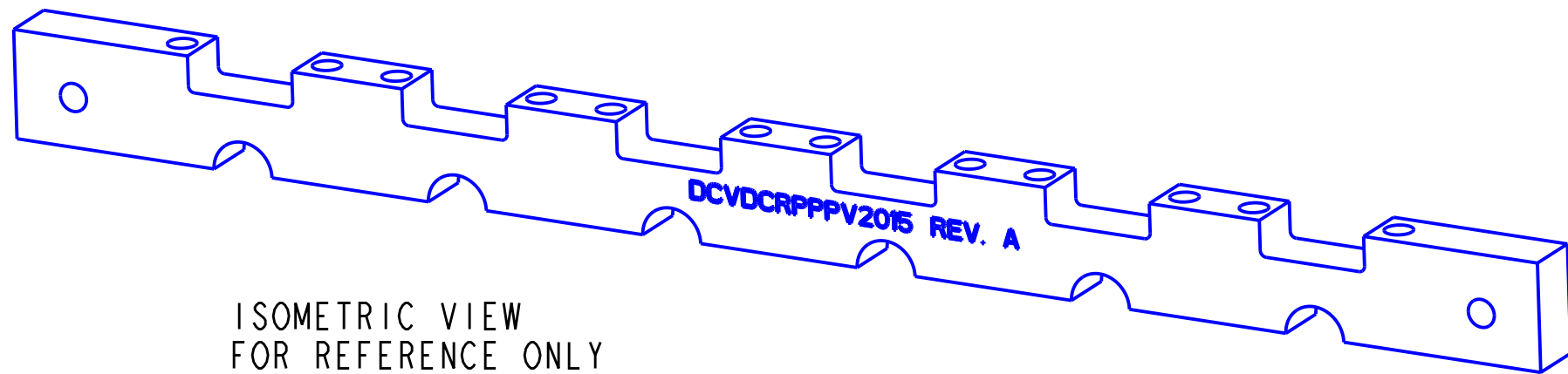
2

1

REVISIONS						
REV	ECN NO.	DESCRIPTION	BY	DATE	CKR	APP
-		INITIAL RELEASE	JF	4/25/2025	MZ	MZ
A		DIM 17.60 WAS 19.05; DIM 8.08 WAS 9.53	JF	2/10/2026	MZ	MZ



SECTION D-D



ISOMETRIC VIEW
FOR REFERENCE ONLY

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Export Administration Regulations 15 CFR Parts 730-774 as EAR99 Technology.

UNTOLERANCED DIMENSIONS ARE BASIC

DIMENSIONAL TOLERANCES

	0.25	[0.010]	A	B	C
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Do Not Scale Drawing

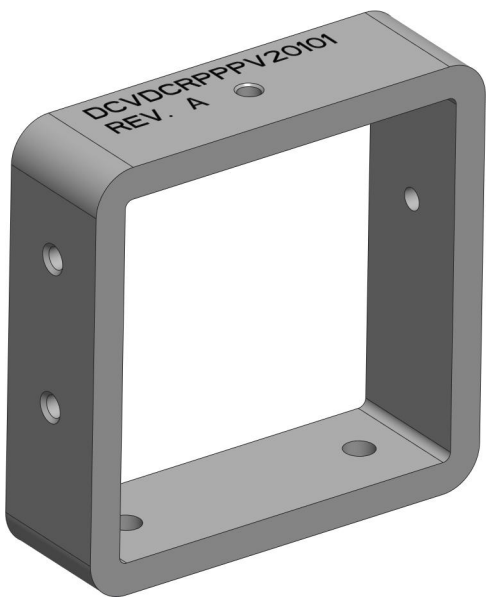
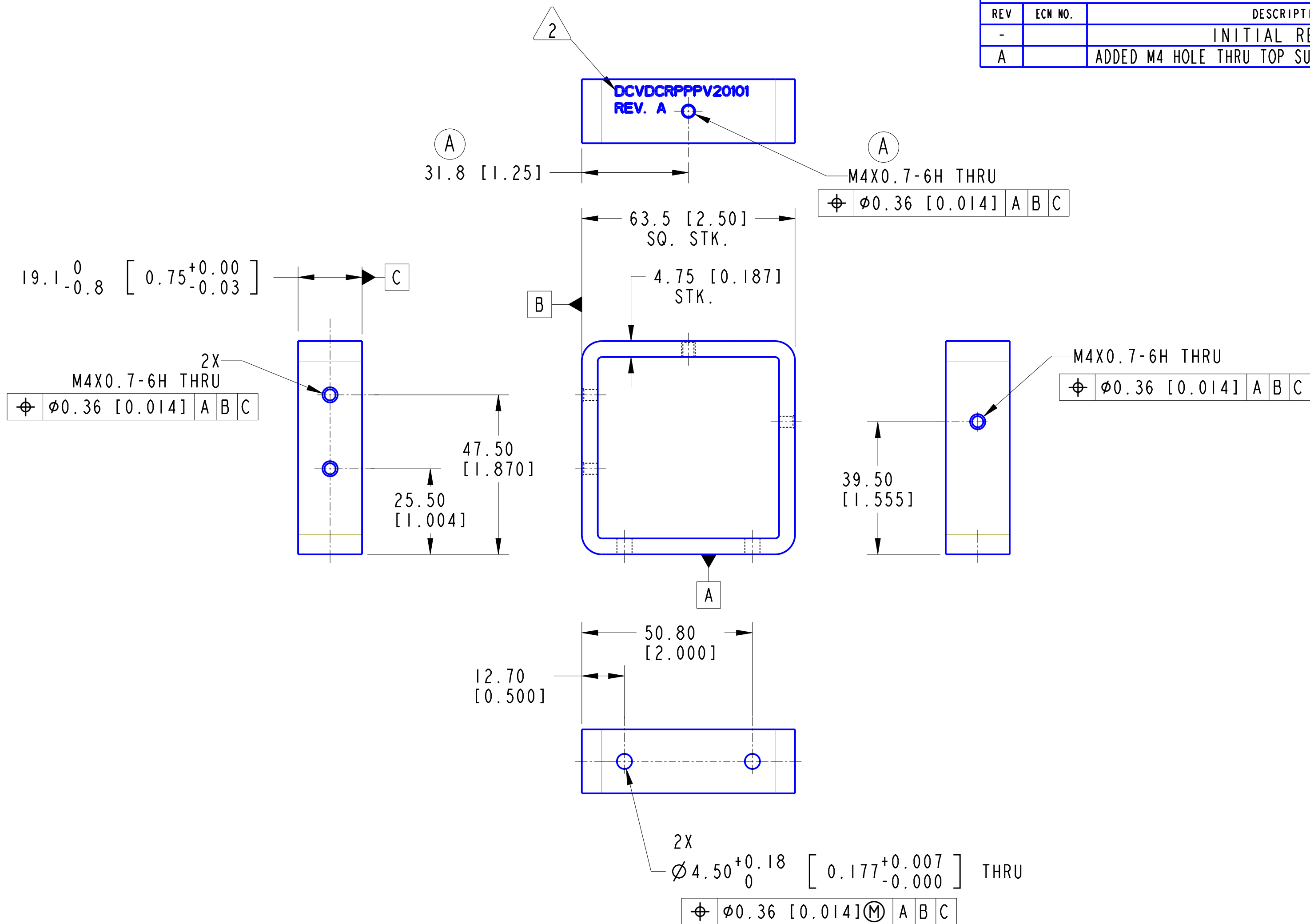
5	THIS DRAWING MAY BE USED WITH MODEL "CLMP_BAR-EDGE CARD-PATCH_PANEL_VD-REV.A.stp" FOR COMPLETE PRODUCT DEFINITION. UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC. VALUES QUERIED FROM 3-D DIGITAL DATA FILE ARE BASIC.
4	PART TO BE FREE OF OILS, GREASE, ETC. PERMISSIBLE TO USE DETERGENTS AND ALCOHOL FOR CLEANING. PART PACKAGING TO MAINTAIN CLEANLINESS.
3	ETCH PART WITH DRAWING NUMBER & REV. LETTER WITH MIN. OF 1.59mm HIGH CHARACTERS IN ACCORDANCE WITH MIL-STD-130. LOCATE APPROXIMATELY WHERE SHOWN.
2	ALUMINUM CHROMATE per MIL-DTL-5541, CLASS 3, CLEAR (RoHS COMPLIANT) Iridite, Alodine, or EQUIVALENT.
1	MATERIAL: 6061-T6 ALUMINUM BAR per ASTM B221, Stock Size: 3/8" Thick x 1" Wide x 12" Lg Suggested Source: McMaster PN: 8975K54
	Notes: Unless otherwise specified:

Do Not Scale Drawing

DCVDCRPPPV202	I
NEXT ASSEMBLY	QTY.
APPLICATION	

DIMENSIONING & TOLERANCING IN ACCORDANCE WITH ASME Y14.5-2009		Brookhaven National Laboratory Physics Department / High-Energy Physics		DEEP UNDERGROUND NEUTRINO EXPERIMENT	
UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC. DIMENSIONS ARE IN MILLIMETERS.		DRAWN BY	J. FARRELL	Apr-25-25	TOP ASSEMBLY:
DIMENSIONS IN BRACKETS [x.xx] ARE IN INCHES AND ARE FOR REFERENCE ONLY.		CHECKED BY	M. ZHAO	4/25/2025	NEXT ASSEMBLY:
DIMENSIONAL TOLERANCES		ENGINEER	M. ZHAO	4/25/2025	DESCRIPTION:
BREAK EDGES & SHARP CORNERS 0.12 [0.005"] MIN. TO 0.75 [0.030"] MAX.		Q.A.	J. PAUCAR	4/30/2025	SIZE:
FINISH 3.2 [125]		WBS#			DRAWING PART NUMBER:
THIRD ANGLE PROJECTION		CLMP_BAR-EDGE CARD-PATCH_PANEL_VD		ESH&Q RISK LEVEL	A-3
				WEIGHT: kg	0.09
				WEIGHT: lbs.	0.193
				SCALE:	1:1
				SHEET:	1/1
				REV.	A

REVISIONS						
REV	ECN NO.	DESCRIPTION	BY	DATE	CKR	APP
-		INITIAL RELEASE	JF	3/6/2025	MZ	MZ
A		ADDED M4 HOLE THRU TOP SURFACE FOR GRND STRAP	JF	2/13/2026	MZ	MZ



ISOMETRIC VIEW
FOR REFERENCE ONLY

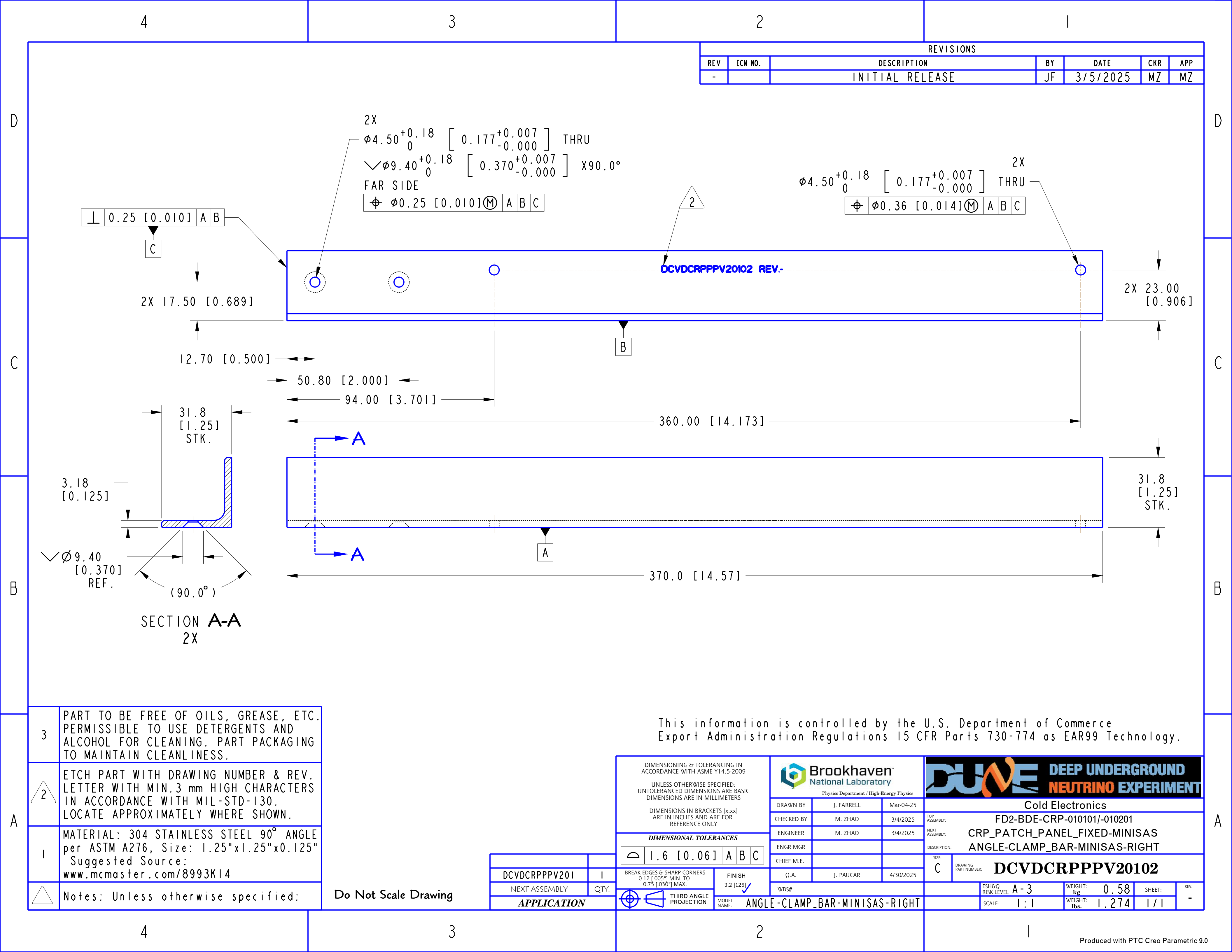
This information is controlled by the U.S. Department of Commerce
Export Administration Regulations 15 CFR Parts 730-774 as EAR99 Technology.

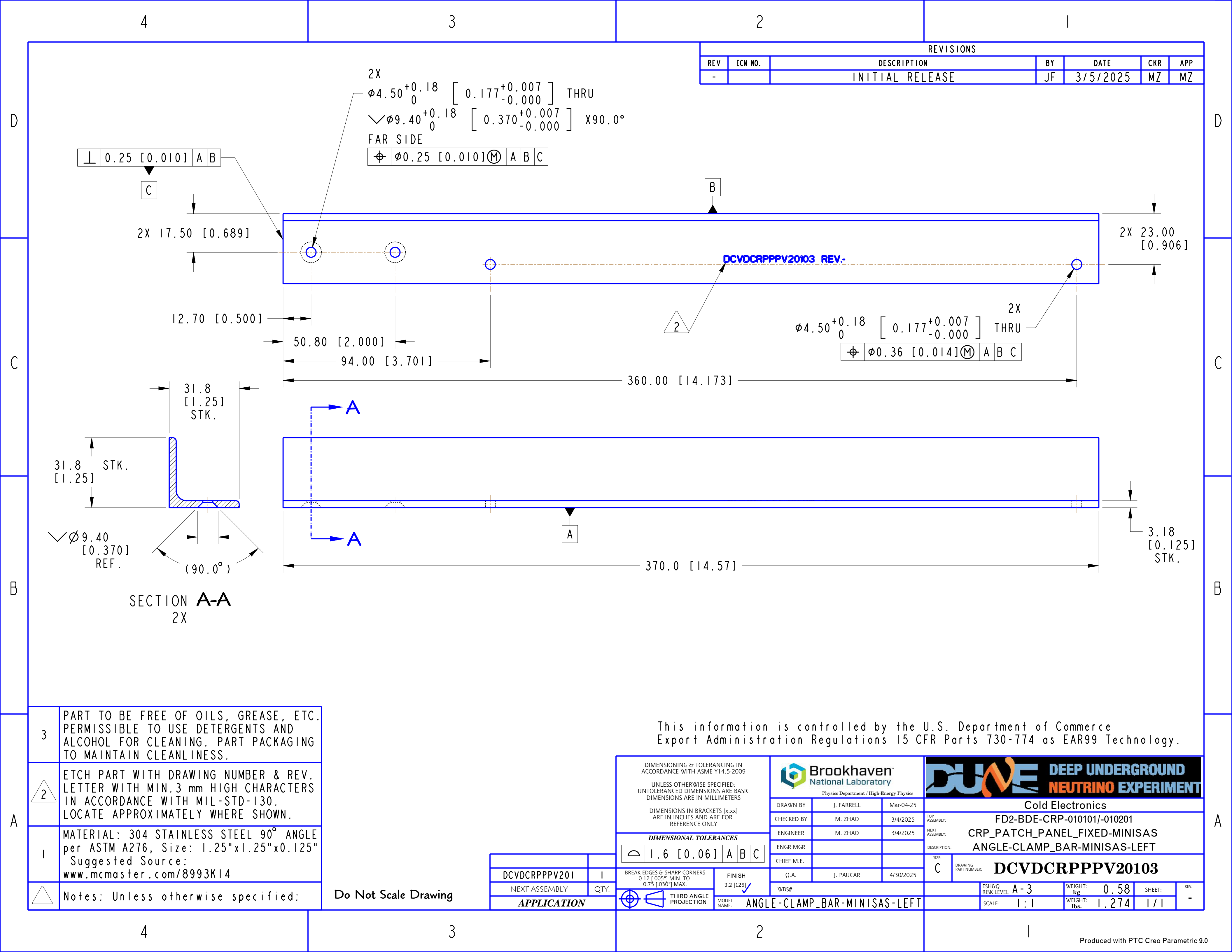
3	PART TO BE FREE OF OILS, GREASE, ETC. PERMISSIBLE TO USE DETERGENTS AND ALCOHOL FOR CLEANING. PART PACKAGING TO MAINTAIN CLEANLINESS.
2	ETCH PART WITH DRAWING NUMBER & REV. LETTER WITH MIN.3 mm HIGH CHARACTERS IN ACCORDANCE WITH MIL-STD-130. LOCATE APPROXIMATELY WHERE SHOWN.
1	MATERIAL: TUBE, SQUARE, 304 SST per ASTM A554 Stock Size: 2.50" Square x 0.187" Wall Suggested Source: McMASTER_89825K88
	Notes: Unless otherwise specified:

Do Not Scale Drawing

DCVDCRPPPV201	2
NEXT ASSEMBLY	QTY.
APPLICATION	

DIMENSIONING & TOLERANCING IN ACCORDANCE WITH ASME Y14.5-2009 UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC DIMENSIONS ARE IN MILLIMETERS DIMENSIONS IN BRACKETS [x.xx] ARE IN INCHES AND ARE FOR REFERENCE ONLY			Brookhaven National Laboratory Physics Department / High-Energy Physics			DUNE DEEP UNDERGROUND NEUTRINO EXPERIMENT		
DRAWN BY J. FARRELL Mar-04-25			CHECKED BY M. ZHAO 3/4/2025			TOP ASSEMBLY: FD2-BDE-CRP-010101/-010201		
ENGINEER M. ZHAO 3/4/2025			ENGR MGR			NEXT ASSEMBLY: CRP_PATCH_PANEL_FIXED-MINISAS		
CHIEF M.E.			Q.A. J. PAUCAR 4/30/2025			DESCRIPTION: TUBE_SQ-CRP_PP_FIXED-MINISAS		
WBS#			SIZE: C			DRAWING PART NUMBER: DCVDCRPPPV20101		
BREAK EDGES & SHARP CORNERS 0.12 [0.005"] MIN. TO 0.75 [0.030"] MAX.			FINISH 3.2 [125]			ESH&Q RISK LEVEL A-3		
THIRD ANGLE PROJECTION			MODEL NAME: TUBE_SQ-CRP_PP_FIXED-MINISAS			SCALE: 1:1		
						WEIGHT: 0.16 kg		
						WEIGHT: 0.362 lbs.		
						SHEET: 1/1		
						REV. A		





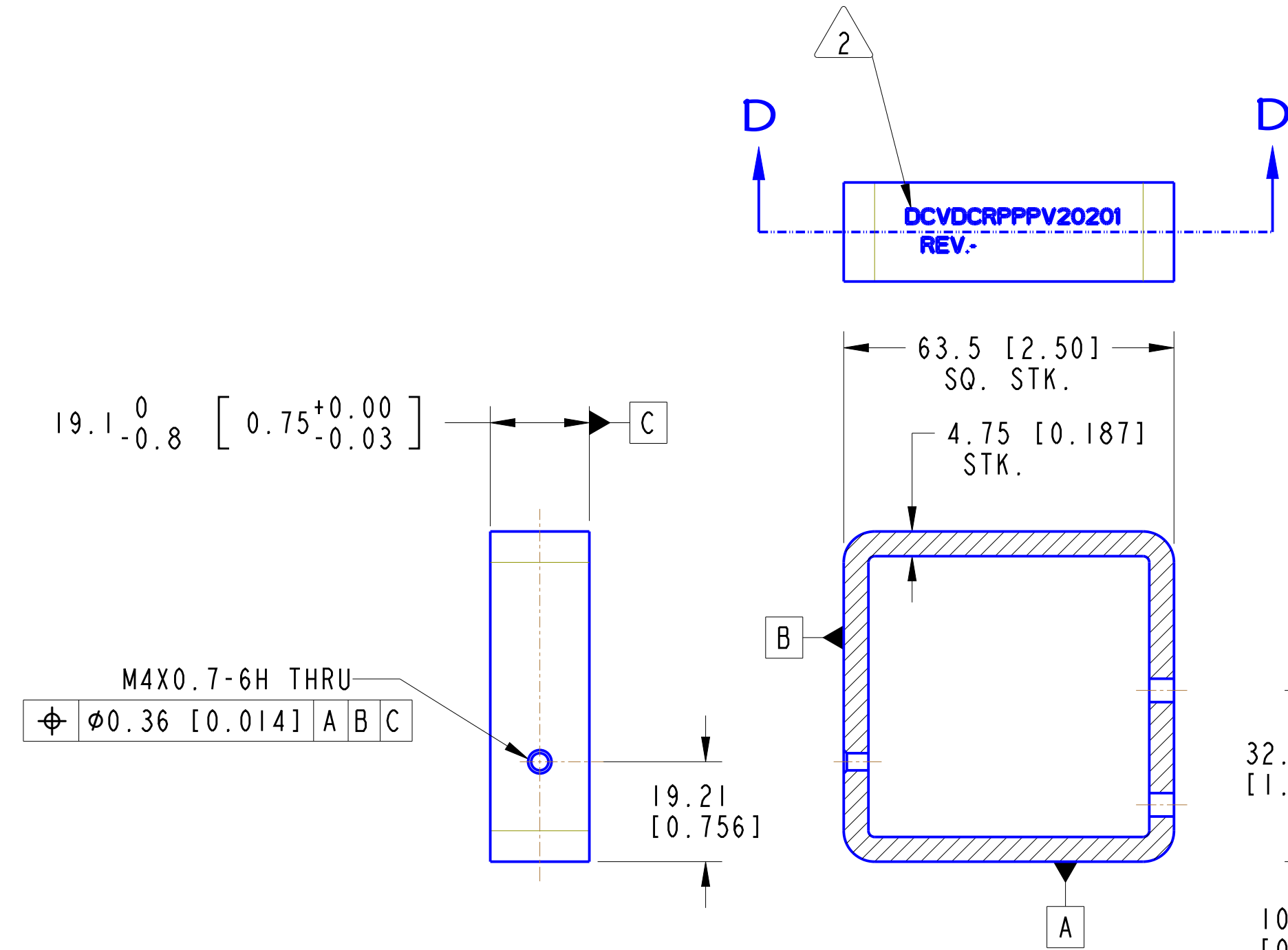
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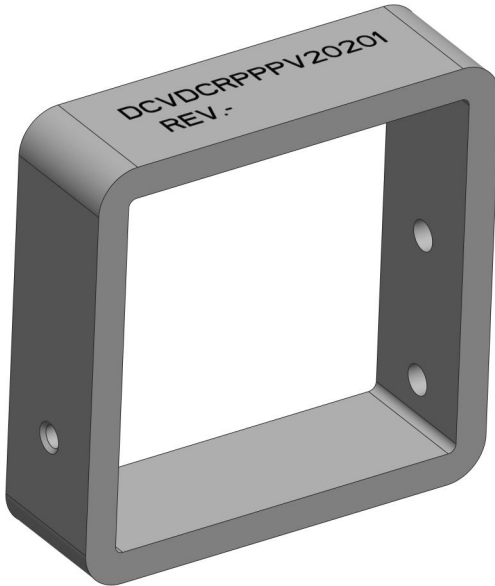
2

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REVISIONS						
REV	ECN NO.	DESCRIPTION			BY	DATE
-		INITIAL RELEASE			JF	3/6/2025
					CKR	APP
					MZ	MZ



SECTION D-D



ISOMETRIC VIEW
FOR REFERENCE ONLY

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Export Administration Regulations 15 CFR Parts 730-774 as EAR99 Technology.

3	PART TO BE FREE OF OILS, GREASE, ETC. PERMISSIBLE TO USE DETERGENTS AND ALCOHOL FOR CLEANING. PART PACKAGING TO MAINTAIN CLEANLINESS.
2	ETCH PART WITH DRAWING NUMBER & REV. LETTER WITH MIN.3 mm HIGH CHARACTERS IN ACCORDANCE WITH MIL-STD-130. LOCATE APPROXIMATELY WHERE SHOWN.
1	MATERIAL: TUBE, SQUARE, 304 SST per ASTM A554 Stock Size: 2.50" Square x 0.187" Wall Suggested Source: McMASTER_89825K88
	Notes: Unless otherwise specified:

Do Not Scale Drawing

DCVDCRPPPV202	2
NEXT ASSEMBLY	QTY.
APPLICATION	

DIMENSIONING & TOLERANCING IN ACCORDANCE WITH ASME Y14.5-2009 UNLESS OTHERWISE SPECIFIED: UNTOLERANCED DIMENSIONS ARE BASIC DIMENSIONS ARE IN MILLIMETERS DIMENSIONS IN BRACKETS [x.xx] ARE IN INCHES AND ARE FOR REFERENCE ONLY			Brookhaven National Laboratory Physics Department / High-Energy Physics			DUNE DEEP UNDERGROUND NEUTRINO EXPERIMENT		
DRAWN BY J. FARRELL Mar-04-25			CHECKED BY M. ZHAO 3/4/2025			Cold Electronics		
ENGINEER M. ZHAO 3/4/2025			ENGR MGR			CRP_PATCH_PANEL_FIXED		
CHIEF M.E.			Q.A. J. PAUCAR 4/30/2025			CRP_PATCH_PANEL_FIXED-EDGE_CARD		
WBS#			SIZE C			TUBE_SQ-CRP_PP_FIXED-EDGE_CARD		
BREAK EDGES & SHARP CORNERS 0.12 [0.005"] MIN. TO 0.75 [0.030"] MAX.			FINISH 3.2 [125]			DRAWING PART NUMBER: DCVDCRPPPV20201		
THIRD ANGLE PROJECTION			MODEL NAME: TUBE_SQ-CRP_PP_FIXED-EDGE_CARD			ESH&Q RISK LEVEL A-3		
						SCALE: 1:1		
						WEIGHT: kg 0.16		
						WEIGHT: lbs. 0.363		
						SHEET: 1/1		
						REV. -		

4

3

2

1

BROOKHAVEN NATIONAL LABORATORY SUPPLIER QUALITY ASSURANCE REQUIREMENTS (BNL-QA-101)

PO/Contract No.: 477155

1.0 PURPOSE & SCOPE

- 1.1 This document establishes quality assurance requirements to which Suppliers to Brookhaven Science Associates (BSA) must conform when specified in the procurement documentation.
- 1.2 This document contains two main sections. Section 3.0 covers the general requirements that are applicable to all Suppliers. Section 4.0 contains special quality requirements that are applicable only when specifically invoked in the procurement documentation.

2.0 DEFINITIONS

- 2.1 The term Procurement documentation means the purchase order (PO), contract, subcontract, Request for Proposal (RFP), Request for Quotation (RFQ) or other written agreement with the Supplier (seller) in which the requirements of BSA are incorporated.
- 2.2 The term Buyer means BSA operating Brookhaven National Laboratory, acting by and through its Procurement & Property Management Division (PPM) issuing the PO/contract.
- 2.3 The term Supplier (seller) means the legal entity, which is the contracting party, with the Buyer with respect to the procurement documentation.
- 2.4 The term article or item means a product and/or a service.

3.0 GENERAL REQUIREMENTS

Unless otherwise specified in the procurement documentation, the following General Requirements apply:

3.1 Supplier's Quality System and Quality Requirements

The Supplier must have and maintain an effective quality system that will, as a minimum, comply with all of the requirements as designated by the following:

☐ **3.1.1** A quality system certified/registered to the ISO 9001 standard: (latest revision as of the date of issuing the procurement documentation).

☐ **3.1.2** A quality system that meets the requirements of the ISO 9001 standard: "Quality Management Systems – Requirements" (latest revision as of the date of issuing the procurement documentation).

☐ **3.1.3** Conformance to Supplier's/Manufacturer's quality program or system.

☒ **3.1.4** Other: Refer to procurement documentation, (e.g. PO, Statement of Work [SOW], specifications, drawings) for quality requirements.

NOTE: Clauses 3.2 through 3.22 apply to all POs and will be included collectively in other procurement documentation when required/specified.

3.2 Assessment by Buyer

The Supplier's quality system is subject to assessments by the Buyer's Representative(s) for conformance with the

requirements of the PO. Supplier or Distributor must allow BSA representatives, BSA customers, and regulatory agencies right of entry into the Supplier's facilities to determine and verify product, processes, records, personnel, material, procedures, and systems.

3.3 Change Approval

No change(s) are permitted to be made to any Buyer requirements, (e.g. part number, model number, etc.) without the prior written approval of the Buyer.

3.4 Responsibility for Subcontractors

It is the responsibility of the Supplier to impose applicable requirements from this document upon their subcontractors. Additionally, the Buyer reserves the right to disapprove, in writing, any subcontractor. For nuclear facility requirements per NQA-1, the Supplier must incorporate NQA-1 requirements in sub-tier procurement documents.

3.5 Responsibility for Conformance

The Supplier is responsible to provide items that conform to the requirements of the PO regardless of any assessments, surveillances, inspections and/or tests by the Buyer or its representatives at either the Supplier's or Buyer's facility. The Buyer reserves the right to request failure analysis and corrective action for non-conforming articles or items submitted or supplied to the Buyer. The Supplier is responsible for notifying the Buyer of any recalls or alerts associated with this PO.

When NQA-1 requirements are imposed, prior to shipment, the Supplier must report any nonconformance that cannot be corrected to full conformance and obtain approval from BNL for possible Use-As-Is or Repair dispositions. Additionally, for commercial grade services, a service performance record must be provided that documents the adequacy of the service performed to the identified critical characteristics documented in the PO/contract.

3.6 Protection of Material and Equipment

The Supplier must employ procedures that assure adequate protection of material and equipment during shipment and while in storage. Such protection must include special environmental packaging, as necessary. All items shipped (originally packaged or repackaged) to BNL or other locations cited in the PO or contract, must comply with the requirements for preservation, packaging and marking as stated in the latest revision of ASTM Standard D 3951 Standard Practice for Commercial Packaging.

3.7 Measuring and Test Equipment (M&TE) Calibration

The Supplier must calibrate any M&TE used in the fulfillment of the PO requirements against certified standards that are traceable to the National Institute of Standards and Technology (NIST), or some other recognized national or international standard, or physical constant. The Supplier must notify the Buyer of any condition found during the calibration, servicing or repair of measuring and test equipment that can affect the end item requirements.

3.8 Suspect Counterfeit Parts

- The Supplier shall verify the procurement source and associated documentation certifying authenticity.
- Appropriate incoming inspection and / or test methods shall be used to detect potential suspect counterfeit items.
- The Supplier shall flow this requirement down to all sub-tier suppliers to prevent the inadvertent use of suspect counterfeit items.
- Distributors shall not modify, rework or repair any items shipped on this order.
- For reference only, a list of authorized/franchised distributors for electronic components can be found at: <https://www.trustedparts.com/>

* For more information refer to the following Department of Energy website: <https://www.energy.gov/ehss/corporate-reporting-analysis/databases/suspectcounterfeit-and-defective-items>

3.9 Electrostatic Discharge Control

Items that are susceptible/sensitive to electrostatic discharge (ESDS) must be handled and packaged to protect them from damage. Items and/or packages must be labeled to indicate the susceptibility to electrostatic discharge.

3.10 Electrical or Fire Protection Equipment, Material, and Systems

All electrical or fire protection equipment, material, and systems delivered to BNL must be certified, listed, or labeled by a Nationally Recognized Testing Laboratory (NRTL). The CE mark is NOT a recognized NRTL certification mark. (For a listing of OSHA-recognized NRTLs, refer to <http://www.osha.gov/>)

For electrical or fire protection equipment, material, and systems which no NRTL accepts, certifies, lists, labels, or otherwise determined to be safe, the Supplier must determine the equipment to be safe for its intended use. The determination must be made on the basis of test data. The determination and test data documents must be made available to BSA prior to or upon delivery for review and acceptance by the applicable BSA Authority Having Jurisdiction (AHJ).

In accordance with 29 CFR 1910.147(c)(2)(iii) whenever new machines or equipment are provided with energy isolating devices, those devices must be designed to accept a lockout device.

3.11 Hoisting & Rigging Equipment

All hoisting & rigging equipment used at BNL must meet the requirements of the latest applicable OSHA Regulations and ASME B30 Series standards for design, construction, markings, and proof load testing.

When proof load testing is required by the standards, a certificate must be provided upon delivery documenting the proof test.

3.12 Deleted – requirements were added to each Purchase Order as the “Advance Notification of Delivery.”

3.13 Powered Machine Shop Equipment

All powered machine shop equipment (e.g., lathe) delivered to BNL must meet the requirements of the latest applicable OSHA 1910 Regulations (e.g., part subpart O). Equipment purchased must include an integrated NFPA 79 compliant emergency stop and an anti-restart device.

3.14 Vehicle-Mounted Elevating and Rotating Aerial Devices

All vehicle-mounted and rotating aerial devices equipment used at BNL must meet the requirements of ANSI A92.2.

3.15 Self-propelled Elevating Work Platform Equipment

All self-propelled elevating work platform equipment (e.g., scissor lift) used at BNL must meet the requirements of ANSI A92.6.

3.16 Manually Propelled Elevating Aerial Platform Equipment

All manually propelled elevating aerial platform equipment used at BNL must meet the requirements of ANSI 92.3.

3.17 Boom Supported Elevating Work Platform Equipment

All boom supported elevating work platform equipment used at BNL must meet the requirements of ANSI 92.5.

3.18 Powered Industrial Trucks and Attachments

Powered industrial trucks (e.g. forklifts, hi-lows) must meet the requirements of the ANSI/ITSDF B56 series for design, construction, markings, and test loading. Industrial truck attachments must be approved by the truck manufacturer and supplied with an attachment data plate indicating the new truck capacities.

3.19 Used Industrial Equipment

For used industrial equipment (e.g. scissor lifts, forklifts, etc.), a Certificate of Conformance as defined in clause 4.16 must be provided as objective evidence and must additionally state that all maintenance and manufacturing alerts have been screened and all required repairs and improvements have been completed. The Supplier must provide records of the last year of maintenance.

3.20 Global Harmonized System Compliance

The supplier of chemicals must deliver the chemical in full compliance with the Department of Labor, Occupational Safety & Health Administration (OSHA)’s Globally Harmonized System (GHS) Hazard Communication Standard (29CFR1910.1200), available at:

<https://www.osha.gov/dsg/hazcom/HCSFinalRegTxt.html>

All hazardous chemicals delivered to BNL must be accompanied by an GHS Safety Data Sheets (SDS) with the format and content specified in 29CFR1910.1200. For information on the GHS SDS see OSHA Brief- Hazard Communication Standard: Safety Data Sheets Publication 3514 available at:

<https://www.osha.gov/Publications/OSHA3514.pdf>

All hazardous chemicals delivered to BNL after 12/01/2015 must have a label with the elements specified in 29CFR1910.1200 [product identifier; pictograms; signal words; hazard statement(s); precautionary statement(s); and manufacturer, importer, or distributor’s name, address, and telephone number]. For information on the GHS label, see OSHA Brief- Hazard Communication Standard: Labels and Pictograms Publication 3636 available at:

<https://www.osha.gov/Publications/OSHA3636.pdf>

3.21 Age/Shelf Life and Storage Control

The Supplier must have an effective storage and age control system for items where acceptability is limited by the age or manner of storage of the item. The system must include a method of identifying the expiration date on the containers in which material is delivered to the Buyer. Special handling conditions must be recorded on certifications and shipping documents covering the material delivered to the Buyer. At the time of receipt, the material must not have less than three-quarters of its shelf life remaining, without prior written approval from the Buyer for each shipment.

3.22 Product Recalls/Product Bulletins/Safety Alerts

Any and all product recall alerts, product bulletins, or safety alerts must be communicated by email directly to ProductRecallAlert@BNL.gov. Provide the Purchase Order

Number(s) and names of purchasers with notification to assist BNL in locating and identifying the subject material.

4.0 SPECIAL REQUIREMENTS

The following Special Requirements are applicable only when specified in the procurement documentation or as indicated by check mark hereon. These Requirements can be modified as required.

INSTRUCTIONS: Since sub-clauses (e.g., 4.4.1) are tied to the main clause (e.g., 4.4), the requirements of the main clause will apply by default whenever any sub-clause is selected (regardless of whether the main clause was selected/checked).

4.1 Quality Assurance Program or Manual

The Supplier must submit a copy of their Quality Assurance Program or Manual with their proposal for review and evaluation.

- ☐ **Suppliers providing A1/A2 but without a quality system compliant with NQA-1, i.e., Commercial Grade Supplier,** must follow applicable Commercial Grade Dedication Instructions as defined in the SOW.

OR

- ☐ Supplier must utilize the QA Program audited and approved by BNL.

4.2 Configuration Control System

The Supplier must establish and maintain a system to assure that all end items (including spares) are of the proper configuration, and that all approved configuration changes are incorporated at the specified effectivity points. Records must be maintained to verify the configuration of each item.

4.3 Process Sheets, Travelers, etc.

The Supplier must maintain a system of process sheets, shop travelers, or equivalent means to define the sequence of manufacturing, inspection, installation and test activities to be performed. Flow sheets, or equivalent, must be provided for sign-off by designated inspection personnel at specified inspection and test points, including, as required, re-inspection and re-test points, to assure completion as well as proper sequencing of required operations.

4.4 Manufacturing/Inspection/Test Plan

Sixty (60) calendar days prior to performance of work, the Supplier must submit for the Buyer's approval a Manufacturing/Inspection/Test Plan for the item(s) to be produced. Once approved, changes/revisions must be approved by the Buyer prior to implementation. The Plan must satisfy one or more of the following as selected:

- ☐ **4.4.1** Identification of parts and subassemblies showing integrated flow into end item(s).
- ☐ **4.4.2** Identification of critical manufacturing operations, as well as inspection and test checkpoints.
- ☐ **4.4.3** The Plan may be a single document, or may make use of existing "travelers," or other suitable planning and control documents.

4.5 "Witness" Points

The Buyer reserves the right to designate selected manufacturing, inspection, and/or test operations as "witness" points. The Supplier must provide the Buyer with five (5) working days' notice in advance of reaching such witness points during the manufacturing and test cycle of each item. For nuclear facility applications (NQA-1), witness points are required when Source Inspection is invoked. (see Clause 4.14)

4.6 Test and Inspection Procedures

Test and inspection procedures required to demonstrate satisfactory completion of requirements must be prepared by the Supplier and submitted to the Buyer for approval sixty (60) calendar days prior to use of such procedures. Once approved, changes/revisions must be approved in writing by the Buyer prior to implementation.

4.7 Special Processes

Processes (e.g., welding, brazing, bonding, plating, chemical machining, chemical coating, chemical cleaning, precision cleaning, heat treating, or waste processing) that either cannot be verified non-destructively or require a unique (special) non-destructive test/inspection (e.g., radiographic inspection, ultrasonic testing, pressure leak testing) must be performed in accordance with detailed written procedures. These procedures must specifically describe the exact manner in which the processes are to be performed. Additionally, the following requirements apply as selected:

- ☐ **4.7.1** Copies of special process procedures must be made available on request for review by the Buyer's representative.

- ☐ **4.7.2** At least sixty (60) calendar days prior to use on items deliverable to the Buyer, the Supplier must submit to the Buyer copies of all applicable process procedures for review and approval. Revisions or changes to Buyer-approved special process procedures must be submitted to the Buyer for review and approval prior to implementation.

4.7.3 Qualification of Procedures, Facilities, Equipment and Personnel

The Supplier must, prior to use, qualify the procedures/specifications, facilities, equipment and personnel that will be used for the performance of special processes. Only those personnel who have been qualified to perform a specific special process must be used to perform that process. Records of such qualification must be available to the Buyer's representative upon request.

4.8 Qualification of Procedures, Facilities, Equipment

Superseded by Sub-clause 4.7.3

4.9 Qualification of Special Process Personnel

Superseded by Sub-clause 4.7.3

4.10 End-Item Documentation Package

The Supplier must provide a documentation package for each shipment of the item(s) supplied, which consists of objective evidence of compliance with PO requirements. This documentation package must be complete, legible, indexed, and traceable to the item supplied. Additionally, the following requirements apply as selected:

- ☐ **4.10.1** Copies of reports of all required or necessary inspections, examinations and tests, properly validated by the Supplier's authorized personnel.
- ☐ **4.10.2** A listing of the as-built configuration of each delivered item; this may be defined by the use of drawing numbers and revisions, unique parts lists or other such means of positive identification.
- ☐ **4.10.3** Copies of nonconformance reports dispositioned as "rework/repair" or "use-as-is", and all BSA approved deviation/waivers.
- ☐ **4.10.4** Copies of material test certificates for specified materials, showing physical and chemical properties.

4.10.5 – Superseded by Clause 4.16

4.11 Release for Shipment

The documentation package required in Clause 4.10 must be approved by the Buyer's representative prior to release of the item for shipment.

4.12 Shipment of Documentation Package to Buyer

Three (3) copies of the documentation package required in Clause 4.10 must be shipped to the Buyer with or prior to each shipment of the purchased items.

4.13 Failure Reporting, Analysis and Corrective Action

The Supplier must maintain a failure reporting, analysis and corrective action system that must, at a minimum, evaluate, analyze and correct failures occurring during qualification, first article and end-item acceptance testing and inspection. The results of all failure evaluations and analyses must be documented and available for review by the Buyer.

4.14 Source Inspection/Verification

Items to be delivered require inspection, tests or surveillance by the Buyer's representative at the Supplier's facility. Five (5) working days advance notice, for acceptance inspections and tests, must be provided by the Supplier to the Buyer to permit scheduling of source inspection. For nuclear facility applications (NQA-1), source verification must be conducted in accordance with a Verification Plan to inspect, test, or examine at predetermined/Witness points (see Clause 4.5). Documented evidence of acceptance must be provided upon completion of all verifications.

4.15 Chemical and Physical Test Report

One copy of the actual chemical and physical test report(s) for each heat, batch or lot must accompany each shipment. Test reports must list the actual parameters tested, the acceptable limits for each parameter, and must contain the actual readings taken during test.

4.16 Certificate of Conformance (C of C)

With each shipment, per the procurement documentation, the Supplier must submit a Certificate of Conformance (C of C). In case of drop shipment, a copy of the certificate must be submitted to the Buyer at the time of shipment. The certificate must identify the purchased item/material/equipment by PO number and quantity, include the title of and be signed by an authorized representative of the company, and must constitute a representation by the Supplier that:

- A. Materials used are those which have been specified by the Buyer, and that the items delivered were produced from materials for which the Supplier has on file, reports of chemical or physical analysis, or any other equivalent evidence of conformance of such items to applicable specifications;
- B. Processes used in the fabrication of items delivered were in compliance with applicable specifications included as part of the PO/contract, or Buyer-approved procedures or specifications;
- C. The items as delivered comply with all applicable drawings, specifications, deviations/waivers and other requirements of the procurement documentation; and-
- D. When specified, cleaning and cleanliness requirements have been completely satisfied. The C of C must reference the Supplier's applicable cleaning procedures.

4.16.1 Additional C of C Requirements for Nuclear (NQA-1) Purchases

A. The certificate must be signed by the person responsible for the quality function and whose function and position is described by the supplier's QA program/system.

B. The Supplier's Certification system must be described in their QA Program/system documentation and include the process for completing the C of C along with the review and Approval of the C of C prior to delivery.

C. The Buyer reserves the right to verify the validity of Supplier's certificates and the effectiveness of the Supplier's certification system. This verification will be commensurate with the Supplier's quality performance/history.

4.17 Report with Each Shipment

Superseded by Clause 4.10

4.18 First Article Acceptance

Buyer acceptance of first article(s) is required prior to the production run. The first article(s) must be identified as such, including the PO number/contract, part number, and part name. The Supplier is required to:

- ☐ **4.18.1** Submit the first article(s) to the Buyer's representative for test/inspection to be conducted at the Supplier's facility by the Buyer's representative.
- ☐ **4.18.2** Submit the first article(s) to the Buyer for test / inspection by the Buyer at the Buyer's facility.
- ☐ **4.18.3** Submit the first article(s) to the Buyer together with documents showing data representing results of the Supplier's first article(s) test/inspection, including the actual dimension or value for each specified characteristic.
- ☐ **4.18.4** After Buyer acceptance of first article(s), all of the remaining units required by the PO/contract must be produced by the Supplier and the Supplier's suppliers using the same design, materials, processes, methods and tooling that were used to manufacture the approved first article(s). Any changes must have prior written approval from the Buyer.

4.19 Notification of Change to Design, Methods, or Processes

The Supplier must immediately notify the Buyer of any significant changes (those that may affect form, fit, function, reliability, safety, or interchangeability) in product design, fabrication methods, materials, or processing from those used by the Supplier at time of Supplier's quotation or offer to the Buyer, which resulted in the PO.

4.20 Age/Shelf Life and Storage Control

Superseded by Clause 3.21

4.21 Serial Numbers

The Supplier must assign/mark a separate and distinct serial number to each end-item in accordance with the procurement documentation. A record of the serial number, for each part number, must be maintained by the Supplier.

4.22 Lot or Batch Numbers

For items furnished in accordance with the procurement documentation, the manufacturing lot or batch number must be indicated on the packing list, certifications and other applicable documents. Where impractical to mark individual parts due to size or shape, the lot or batch number must be marked on identifying tags or the smallest unit package.

4.23 Material Traceability

Materials used must be identified by material type, applicable specification and revision number, and be traceable to their lot and/or heat number(s). Traceability records must be available for review by the Buyer's representative.

4.24 Shipment Destination Other than BNL

The material ordered is to be shipped to other than the Buyer's facilities. Copies of the data required in accordance with the procurement documentation must accompany the shipment; in addition, one copy of such data must be mailed to the Buyer on the same day that shipment is made.

4.25 Heat Treat Bars

Superseded by Clause 4.7

4.26 Burn-in

Burn-in must be performed on each completed item, per the procurement specification or Supplier's Burn-In process approved by the Buyer. Records of burn-in testing, repairs and test results must be maintained and must be available to the Buyer's representative upon request.

4.27 Welding Procedures

Superseded by Clause 4.7

4.28 Weld/Braze Inspection Report

A report(s) must be submitted that indicates the complete inspection of welds or brazes from the initial fit-up stage through final inspection. Inspection reports must be accompanied by all radiographic films, filler metal reports etc. The reports must contain the signature or stamp, and title of an authorized Supplier representative.

4.29 Radiographic Quality Requirements

Items requiring radiographic inspection must be radiographed and processed in accordance with the Supplier's special process procedures that satisfy design specifications, standards or other procurement documentation requirements. Personnel reading and interpreting film must have been examined and certified. Responsibility for this certification must rest with the Supplier, whether the Supplier does the work or subcontracts to a specialized laboratory. A report of the findings must include the name of the reader and the signature and title of a responsible representative. The radiographic film and a reproducible copy of the report must accompany each shipment. An adequate method of identifying and cross-referencing each film exposure, report, and item must be provided. When parts are serialized, serial numbers must appear on the report and the film.

4.30 Nondestructive Test Reports

All nondestructive testing must be conducted in compliance with the Supplier's special process procedures that satisfy the applicable provisions of the design specifications, or other procurement documentation requirements. Personnel and equipment utilized in performance of such tests must be qualified for the type of test performed. The Supplier must furnish with, or prior to, each shipment reports of such nondestructive examination of material or items furnished. These reports must be identifiable to the respective item or material including the specific section, joints or views of the item furnished. These reports must contain the signature and title of an authorized Supplier representative. When items are serialized, the serial numbers must appear on the reports.

4.31 Pressure or Leak Test Reports

Test reports must be prepared for all pressure and leak tests. Such reports must state the requirement, the Supplier's test procedure

number, and the observed result for each item, joint or connection tested. When items are serialized, the serial numbers must appear on the report. Reports must contain the signature/title of an authorized Supplier representative and must accompany each shipment.

4.32 Cleaning Certification

Superseded by Clause 4.16 D

4.33 Calibration Certification

The Supplier must submit with each instrument/system a certification that the instrument/system has been calibrated and is ready for use. The certification must contain, at a minimum, the identity of the instrument/system, identification of the calibration procedure used, identification of the standards and/or equipment utilized for the calibration, and a statement that the calibration of the standards and/or equipment used is traceable to the NIST or some other recognized national or international standard, or physical constant. Unless otherwise specified, detailed support data must remain on file for minimum of three (3) years with the Supplier and must be available for review by the Buyer. The certification must also contain the signature and title of an authorized Supplier representative.

4.33.1 The Supplier will provide "As Found" (i.e., before) and "As Left" (i.e., after) measurements with the certification.

4.33.2 Additional NQA-1 Calibration Certification Requirements

The certificate must be signed by the person responsible for the quality function and whose function and position is described by the supplier's QA program/system.

The Supplier's Certification system must be described in their QA Program/system documentation and include the process for completing the C of C along with the review and Approval of the C of C prior to delivery.

The Buyer reserves the right to verify the validity of Supplier's certificates and the effectiveness of the Supplier's certification system. This verification will be commensurate with the Supplier's quality performance/history.

The certificate must include the following:

- M&TE ID or Asset number
- The Supplier's calibration procedure
- The recommended frequency of calibration
- The calibration date
- The next calibration due date
- As-found calibration data must be reported in the certificate of calibration when calibrated items are found to be out-of-tolerance.
- The standards used for calibration
- The PO number
- Any additional technical and quality requirements, as necessary.

On the certificate, the supplier must also attest that the contracted calibration or test service has been performed in accordance with their ISO/IEC-17025 program, has been performed within their scope of accreditation, is traceable to NIST, the PO's requirements are met, and identify any and all nonconforming conditions.

4.34 Operating-Maintenance Manual

Documentation containing operating procedures, maintenance instructions, spare parts lists, and handling procedures must be submitted with the shipment of the first item.

4.35 Computer Software Configuration Management

The Supplier must have and maintain an effective software configuration management system. The Supplier's system must establish requirements for placing software under configuration control, provide for the positive identification of software, and the control of all software baseline changes.

- ☐ **4.35.1** The Supplier must submit a copy of their software configuration management procedure(s) with their proposal for review and evaluation, and specifications for software features, including requirements for safety, security, functions, and performance.

4.35.2 Superseded by Sub-clause 4.35.1

4.36 Computer Software Verification and Validation (V&V)

The Supplier must develop written procedures describing the controls applied to the design of software and the validation of the design through independent technical review. The procedures must provide for documentation of review activities, including requirements for documenting comments and resolution of comments. Supplier software designs and review documentation must be subject to review and approval by the Buyer.

The Supplier must test and verify computer software developed or modified to fulfill the requirements in the procurement documentation. The verification testing must be accomplished by a comparison of test results with those from other verified software, or by a comparison with results from analytical solutions or Buyer-approved alternatives.

4.37 Computer Software Notification of defects, new releases or other issues

For Safety Software, the Supplier must provide a process to (a) notify users of defects, new releases, or other issues that impact the operation and (b) mechanisms for the users of the software to report defects and request assistance in operating software.

4.38 Computer Software User Training

The Supplier must provide formal on-site, off-site or on-line training for users.

4.39 Records

The Supplier must retain objective evidence, including records, of the inspections and tests performed in the course of manufacturing, testing, inspecting, preserving, packaging, and preparation for shipment of procured items. These records must be made available to the Buyer's representative for review upon request. These records must be maintained for a minimum of three (3) years, unless otherwise specified in the procurement documentation, after the completion of the PO/contract.

4.40 Electrical, Fire Protection, or Scaffolding Equipment, Material, and Systems

Superseded by Clause 3.10

4.41 Hoisting & Rigging Equipment

Superseded by Clause 3.11

4.41.1 Powered Industrial Truck Attachments

Superseded by Clause 3.18

4.41.2 Custom-made Equipment

Superseded by Clause 3.11

4.41.3 Critical Lifts

Superseded by Clause 3.11

4.42 Marking of Outer Package and Hoisting & Rigging Services

Superseded by Clause 3.12

4.43 Franchised/Licensed "Distributor" Traceability

Products that are not purchased directly from the Original Equipment Manufacturer (OEM)/Original Component Manufacturer (OCM) must be purchased only from a franchised/licensed distributor of the product being offered. The distributor must provide documentation that the distributor is contractually authorized by the manufacturer as a franchised / licensed distributor along with the geographic region(s) covered by the authorization. The distributor must not use brokers (any company, person, or entity who is not an OEM/OCM or authorized by the OEM/OCM) for the purchase of items, unless pre-approval has been granted by Brookhaven Science Associates (BSA). For reference only, a list of authorized/franchised distributors for electronic components can be found at:

<https://www.trustedparts.com/>

- ☐ **4.43.1** The distributor must ensure traceability of all products to the Original Equipment Manufacturer (OEM)/Original Component Manufacturer (OCM) by identifying the Original Equipment Manufacturer (OEM)/Original Component Manufacturer (OCM) for each lot and / or date code on the Certificate of Conformance. The distributor must provide a copy of the Manufacturer's certificate for the lot number being supplied, along with their franchised distributor documentation.

4.44 Power Machine Shop Equipment

Superseded by Clause 3.13

4.45 Aerial Lifts Equipment

Superseded by Clause 3.14

4.46 Self-propelled Elevating Work Platform Equipment

Superseded by Clause 3.15

4.47 Nuclear (NQA-1) Calibration Services

The service must be provided in accordance with the supplier's accredited ISO/IEC-17025 program and scope of accreditation. As-found calibration data must be reported in the certificate of calibration when calibrated items are found to be out-of-tolerance. The equipment and standards used to perform the calibration must be identified in the certificate of calibration. Subcontracting of these accredited services is prohibited. BNL must be notified of any condition that adversely impacts the supplier's ability to maintain the scope of accreditation. Performance of the services listed on this order is contingent on the Supplier's accreditation having been achieved through an on-site accreditation assessment by the accreditation body within the past 48 months. Any additional technical and quality requirements, as necessary.

4.48 Nuclear (NQA-1) Right of Access

BNL or designated representative may enter the premises of Supplier (or its sub-tier supplier) and review records for the purpose of inspection, surveillance, or quality assurance audit.

4.49 Nuclear (NQA-1) Partial Orders

Supplier may not ship partial orders without prior written approval from BNL.